

EECS 836: Machine Learning

Zijun Yao

Assistant Professor, EECS Department
The University of Kansas

Class and Office Hour

- **Instructor:** Zijun Yao, Eaton Hall 2048 (Office)
- **Class:** 4:00pm - 5:15pm Monday/Friday, LEEP2 2300
- **Office hour:** 12:00pm - 1:00pm Monday or by appointments
- **E-mail:** zyao@ku.edu
 - Recommended Subject: **EECS836 <Your Last Name> <Brief headline>**
- **Course web:** <https://canvas.ku.edu/>

Course Coverage

- Fundamental concepts in machine learning
- Deep neural networks (MLP, CNN, RNN, Transformer, GenAI)
 - How to train deep learning models with popular deep learning frameworks such as PyTorch
- How machine learning is used in real-world application
 - Hands-on (e.g., computer vision, natural language processing, recommender systems)
 - Analyzing business cases

Syllabus and Course Schedule

- Course schedule available under [Syllabus](#) section at Canvas
- Lecture slides and tutorial will be posted under [Lecture Slides](#) section at Canvas
- Assignments, exams and other course work will be posted under [Assignments](#) section at Canvas
- You should check [Announcements](#) frequently to remain updated.

Course Components

- Lectures on Monday and Friday
 - Show up and actively engage
- Slides
 - Slides will be posted before lecture
- Readings
 - Read material before the class
- Assignments and exams
 - Submit assignments on time.
- Project
 - Apply ML algorithms on real-world data and present the results

Grading Policy

1.	Attendance	4%
2.	Assignments	26%
3.	Project	20%
4.	Exam I	25%
5.	Exam II	25%
Total		100%

- **Attendance:** Check 4 times by random
- **Assignments:** Late submissions will receive less credit
- **Exams:** There will be no make-up exams
- **Team project:** a team with at least 3 students, up to 4; proposal and report will be required.
- The final grade is based on a curve
- Active class participation earns up to 3% credit
- **Academic integrity:** Do **NOT** cheat in any homework and exam. Highly identical answer will require explanation.

Project

1. **Group.** Up to 4 students.
2. **Proposal.** Maximum 8 PPT pages of project proposal, dataset, problem definition, data processing, models and (optional) expected outcomes.
3. **Report.** Maximum 15-page report (single-space, 12-point font) consists of sections of introduction, motivation, method, results, and conclusion.
4. **Project tutorial.** Will be available in following lectures

Project

- **Topic:** anything you're interested (sentiment analysis, recommendation systems, healthcare, Covid-19, finance, security)
 - Search on Internet website for ideas
 - Kaggle has rich resources including ideas, data and code
<https://www.kaggle.com/>

- Define input-output



Dog

- **Scope:** not too broad, not too narrow (eg., training a linear classifier for standard dataset)
- **Evaluation metric:** quantitative measure or insights

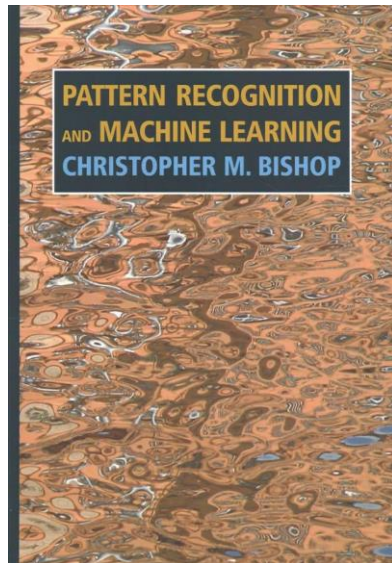
Will discuss more in project tutorial

Course Prerequisites

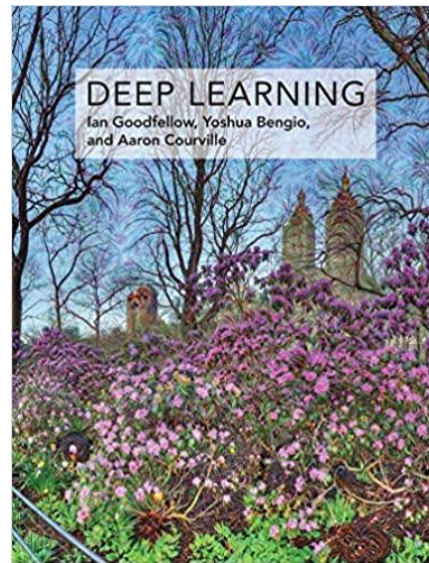
- **Algorithm skills** - basic algorithms and data structures in python (EECS 168 Programming I or equivalents)
- **Math skills** - linear algebra, probability, and statistics (MATH 526 Applied Mathematical Statistics I or equivalents)
- **Programming** with Python

Textbooks (Optional)

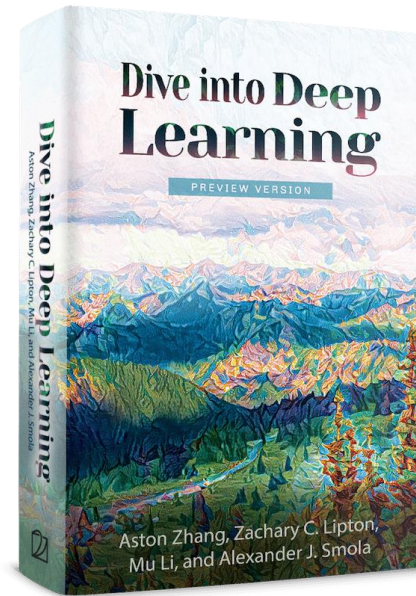
- Use following books as the primary reference



available at
<https://www.micrsoft.com/en-us/research/people/cmbishop/prml-book/>



available at
<https://www.deeplearningbook.org/>



available at
<https://d2l.ai/>

CS229 Lecture notes

Andrew Ng

Part V Support Vector Machines

This set of notes presents the Support Vector Machine (SVM) learning algorithm. SVMs are among the best (and many believe is indeed the best) "off-the-shelf" supervised learning algorithm. To tell the SVM story, we'll need to first talk about margins and the idea of separating data with a large "gap." Next, we'll talk about the optimal margin classifier, which will lead us into a digression on Lagrange duality. We'll also see kernels, which give a way to apply SVMs efficiently in very high dimensional (such as infinite-dimensional) feature spaces, and finally, we'll close off the story with the SMO algorithm, which gives an efficient implementation of SVMs.

1 Margins: Intuition

We'll start our story on SVMs by talking about margins. This section will give the intuitions about margins and about the "confidence" of our predictions; these ideas will be made formal in Section 3.

Consider logistic regression, where the probability $p(y = 1|x; \theta)$ is modeled by $h_\theta(x) = g(\theta^T x)$. We would then predict "1" on an input x if and only if $h_\theta(x) \geq 0.5$, or equivalently, if and only if $\theta^T x \geq 0$. Consider a positive training example ($y = 1$). The larger $\theta^T x$ is, the larger also is $h_\theta(x) = p(y = 1|x; \theta)$, and thus also the higher our degree of "confidence" that the label is 1. Thus, informally we can think of our prediction as being a very confident one that $y = 1$ if $\theta^T x \gg 0$. Similarly, we think of logistic regression as making a very confident prediction of $y = 0$ if $\theta^T x \ll 0$. Given a training set, again informally it seems that we'd have found a good fit to the training data if we can find θ so that $\theta^T x^{(i)} \gg 0$ whenever $y^{(i)} = 1$, and

1

available at
<https://github.com/maxim5/cs229-2018-autumn/tree/main/notes>

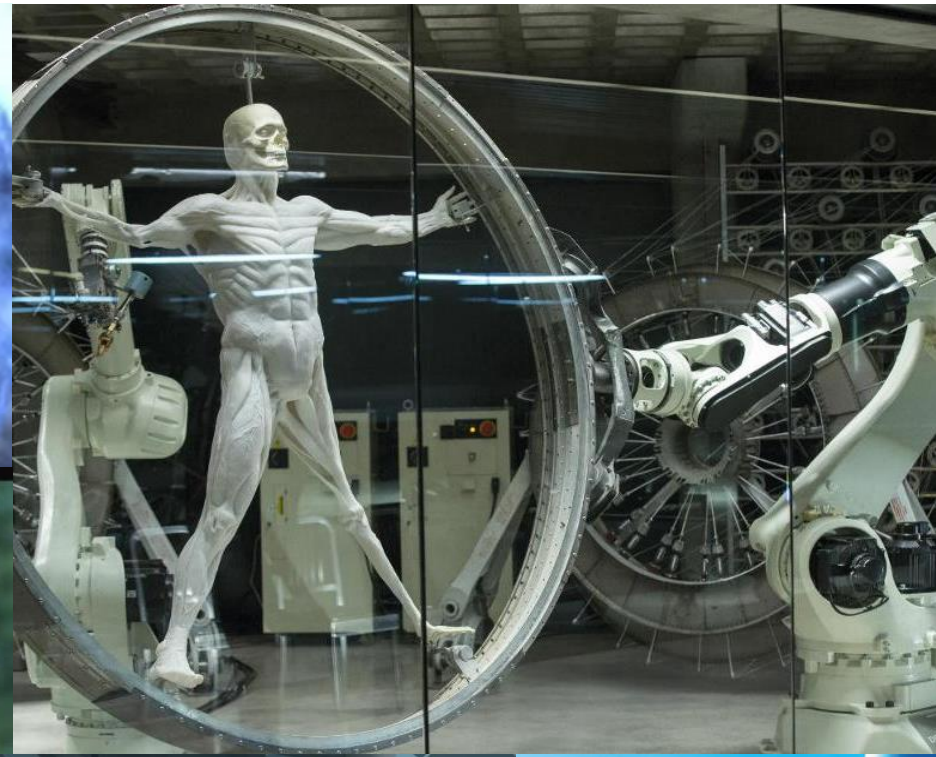
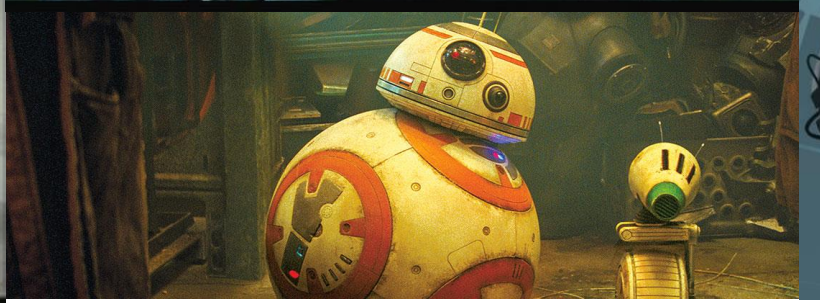
Prepare Yourself for the Future

- AI will create 133 million new and displace 75 million old jobs worldwide (with the net creation of 58 million of new jobs) within the next few years (According to a World Economic Forum report)
- Contributing up to \$15 trillion to the global GDP by 2030, according to PwC.
- There is an acute AI skills shortage around the world: the demand for the AI jobs is measured in millions

Course Expectation

- Develop a strong vocabulary and understanding of ML techniques
- Make informed trade-offs on what ML approaches to use
- Communicate with confidence among developers and consultants
- Hands-on experience via a staged progression of exercises using application data
- Exposure to various AI/ML tools
- Approach business/research problems analytically by identifying opportunities to derive actionable insights from data

What Do You Think of AI



Robotics



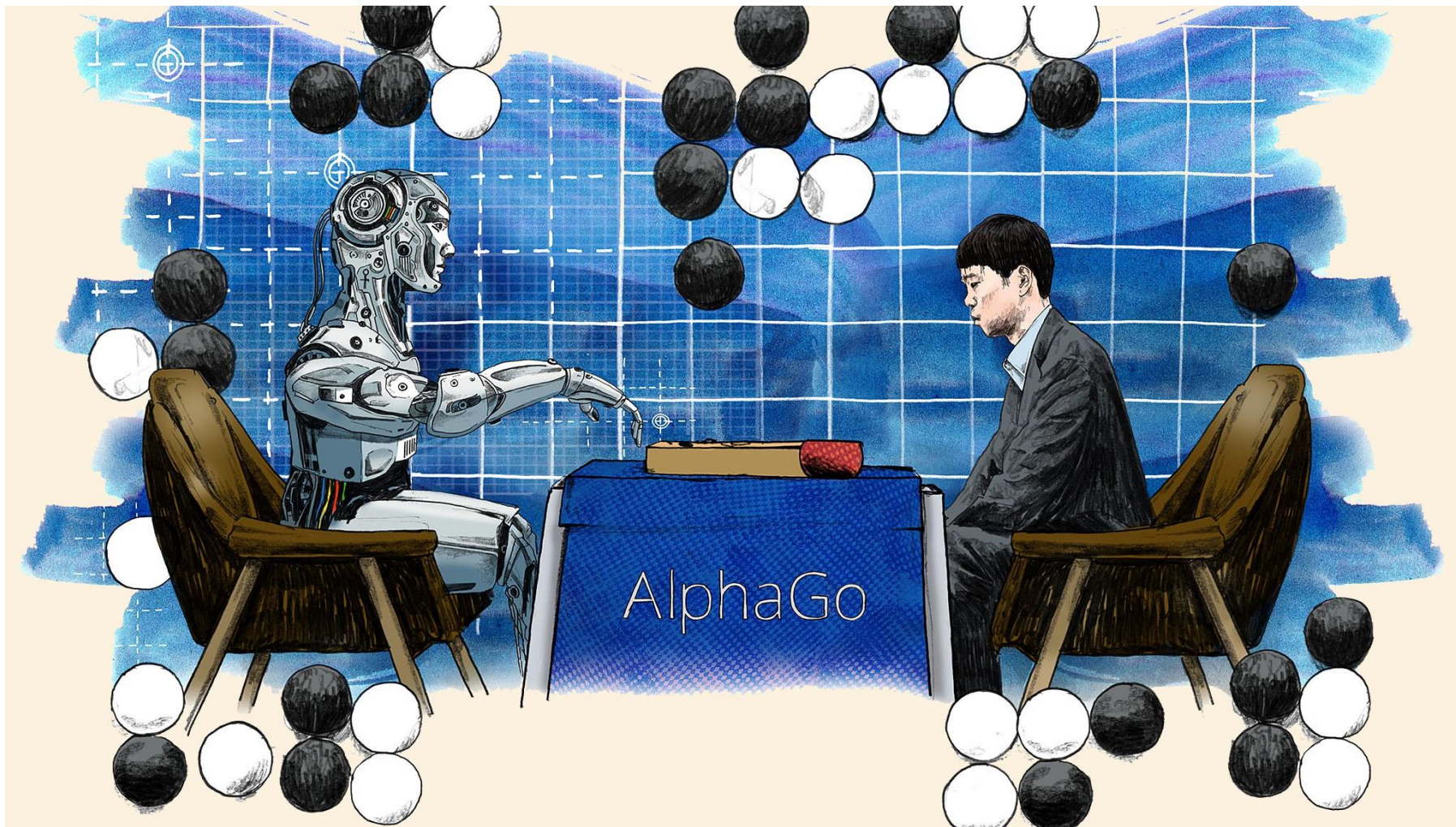
<https://www.youtube.com/watch?v=fn3KWM1kuAw>

Autonomous cars



<https://www.youtube.com/watch?v=tIThdr3O5Qo>

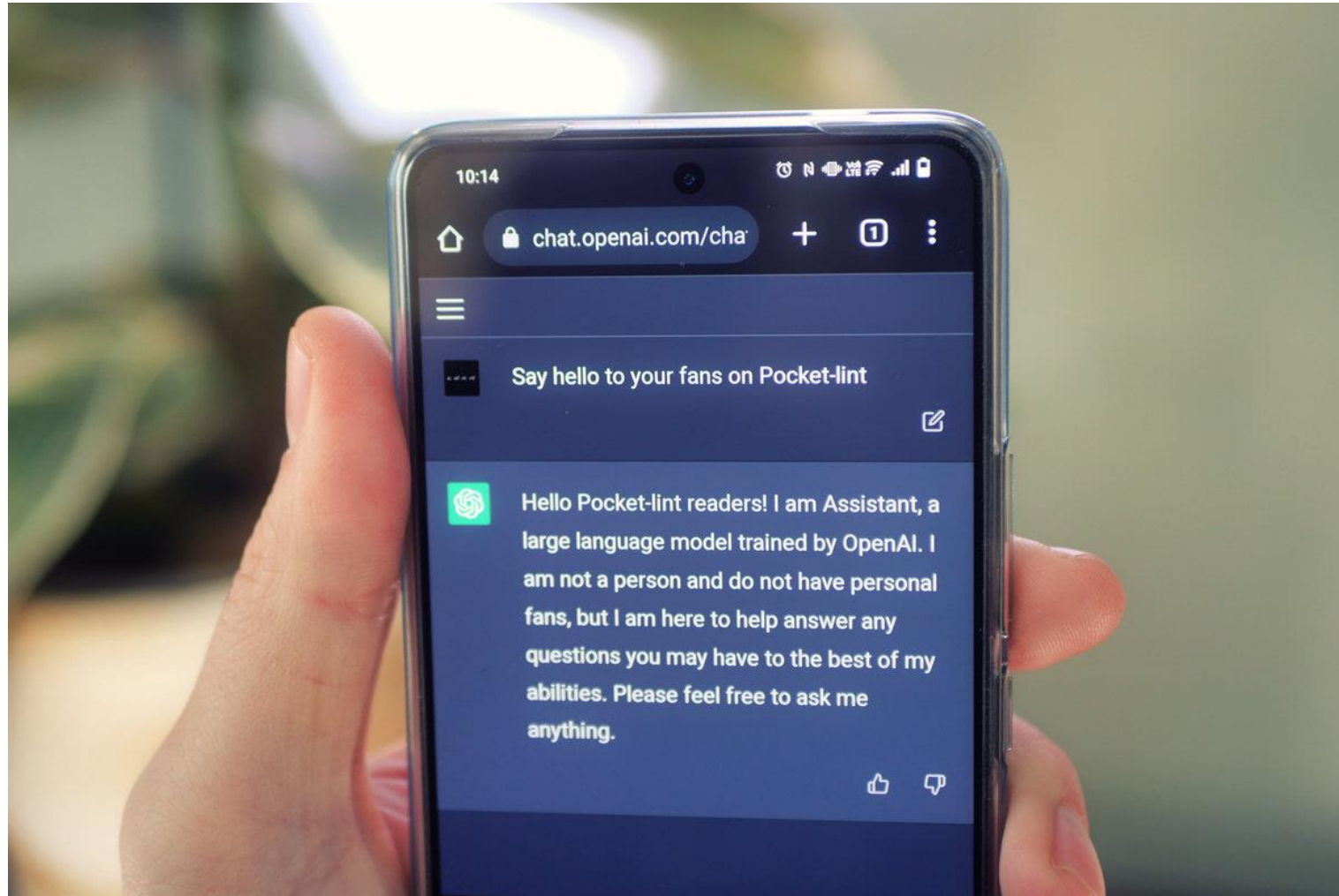
Alpha Go



AlphaGo - The Movie | Full Documentary

<https://www.youtube.com/watch?v=WXuK6gekU1Y>

ChatGPT



It's Time to Pay Attention to A.I. (ChatGPT and Beyond)
<https://www.youtube.com/watch?v=0uQqMxXoNVs>

Stable Diffusion



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$> gta jim halpert john  
krasinski profile picture by  
greg rutkowski, dynamic pose,  
intricate, futuristic,  
fantasy, elegant, by stanley  
artgerm lau, greg rutkowski,  
thomas kindkade, alphonse  
much, loish, norman  
rockwell, fantasy lut,  
asymmetric, long hair, retro  
computer graphics, video  
game, fluid lines
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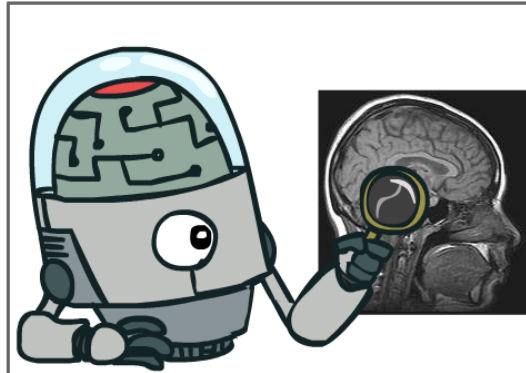


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game, fluid lines
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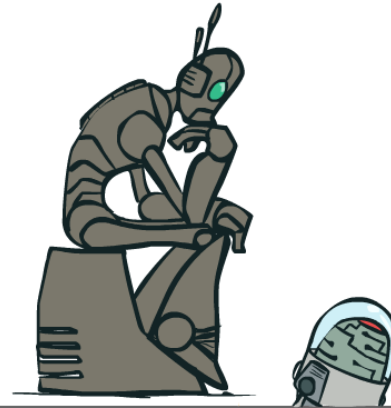
What is AI?

The science of making machines (or computers) that

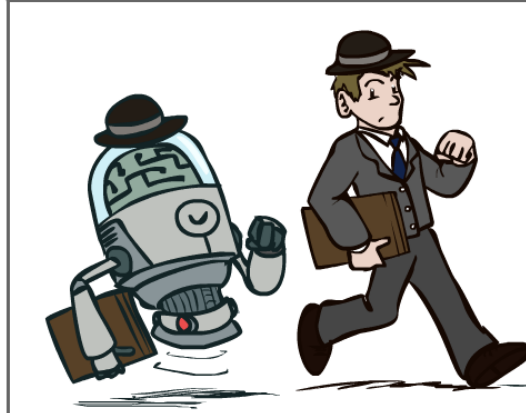
Think like
people



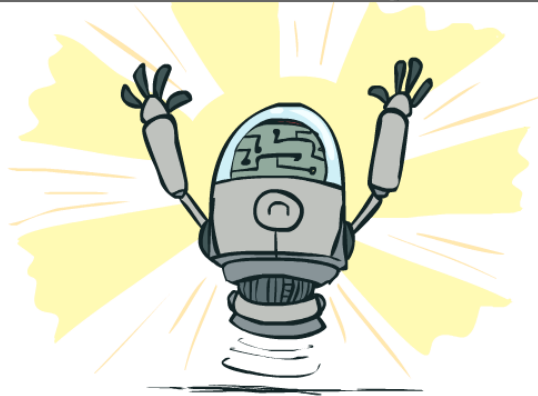
Think
rationally



Act like
people



Act
rationally



AI - Turing test (1950)



No. 236.]

[October, 1950

MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY

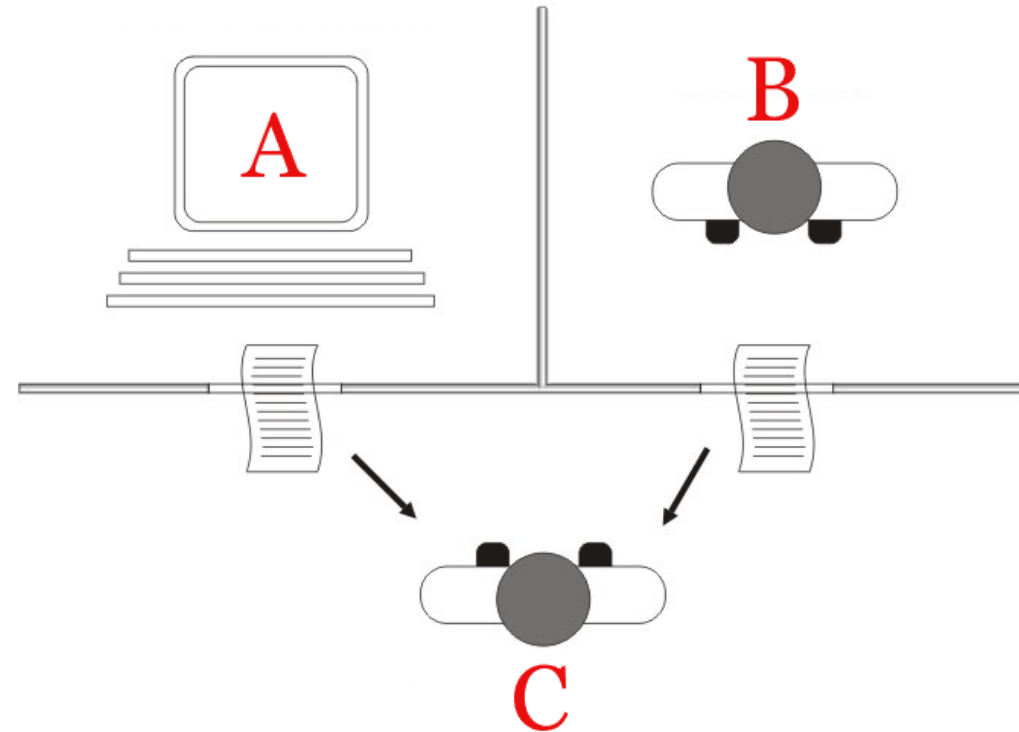


I.—COMPUTING MACHINERY AND INTELLIGENCE

BY A. M. TURING

1. *The Imitation Game.*

I PROPOSE to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to

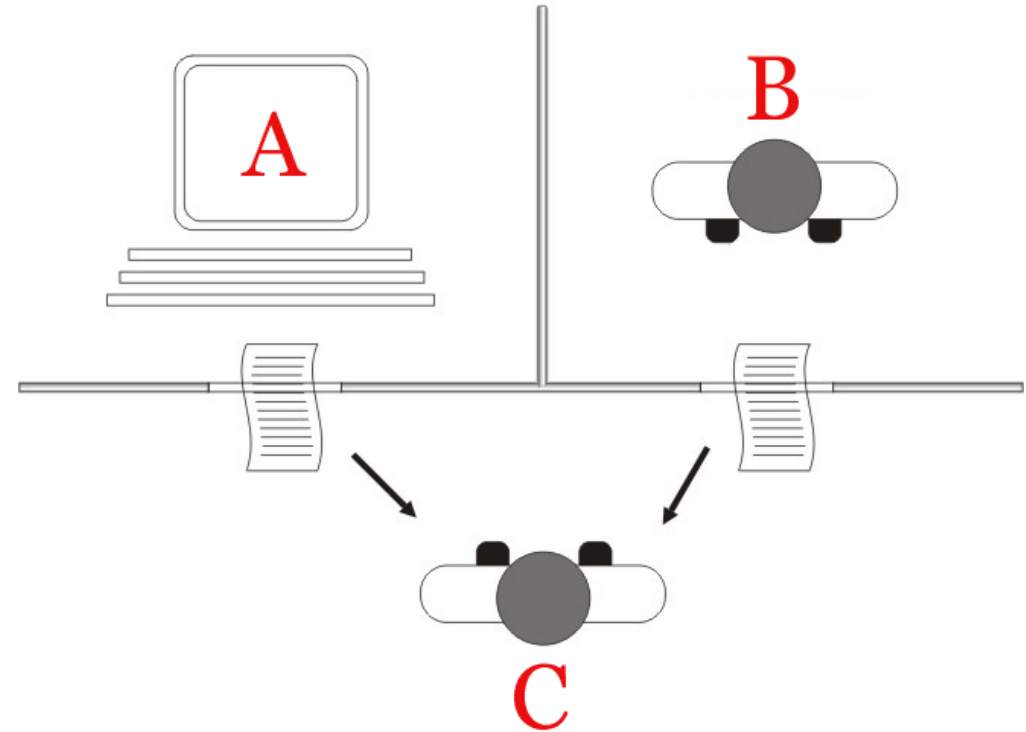


Turing test: A computer can be said to be intelligent, or “think”, if a human judge cannot tell if he/she is interacting with a human or a machine

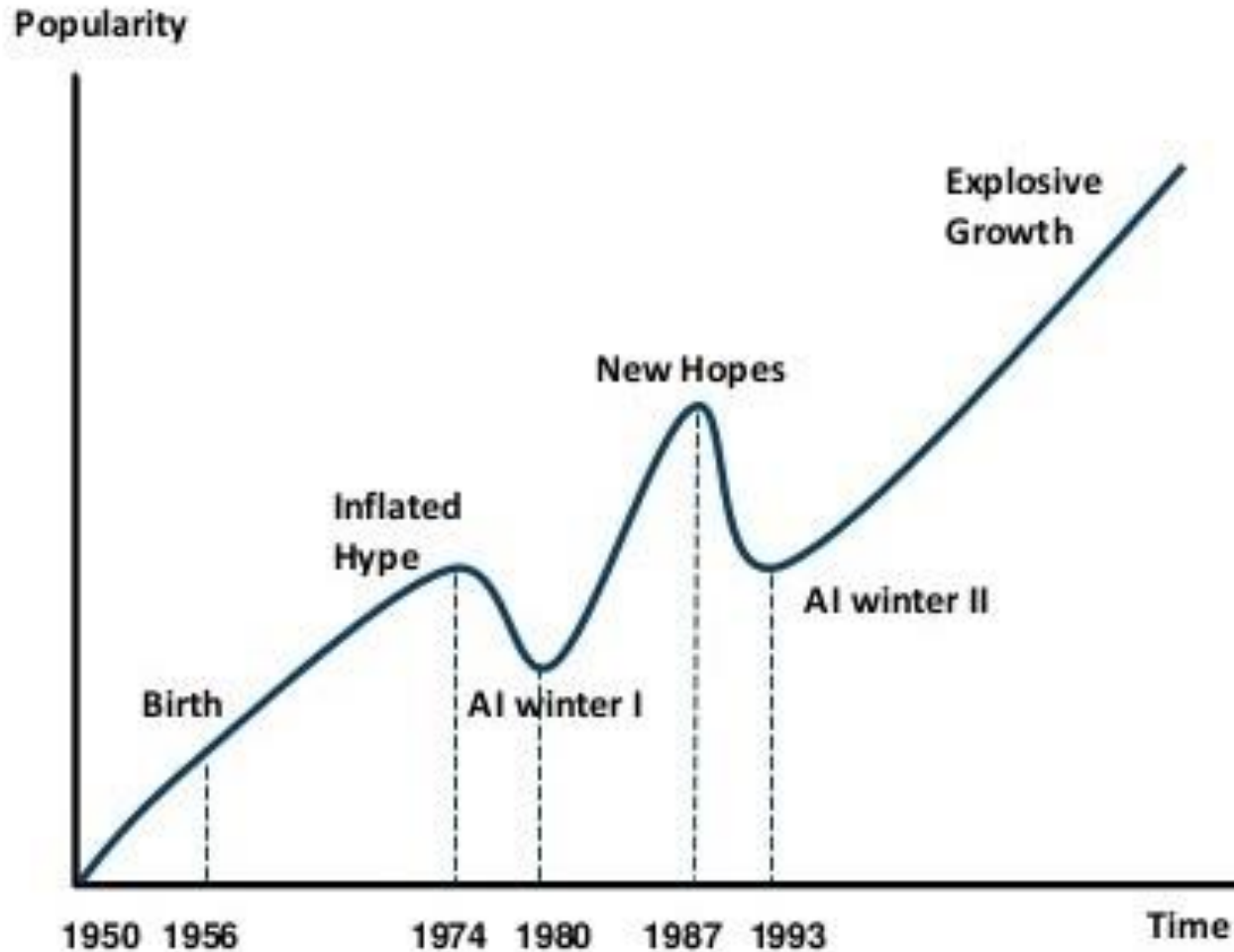
AI - Turing Test

The computer needs following capabilities to pass the test

- Natural language processing
- Knowledge representation
- Automated reasoning
- Machine learning
- Computer vision
- Robotics



Machine Learning: Since 1990s



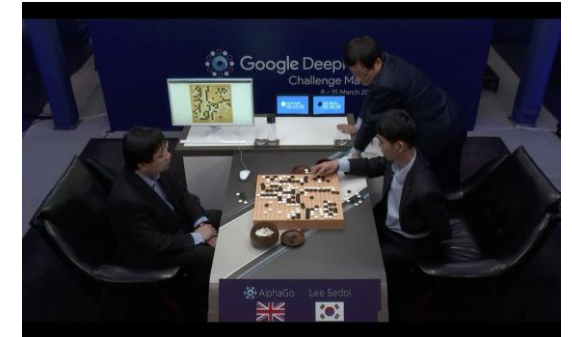
History of AI

- **Machine learning**

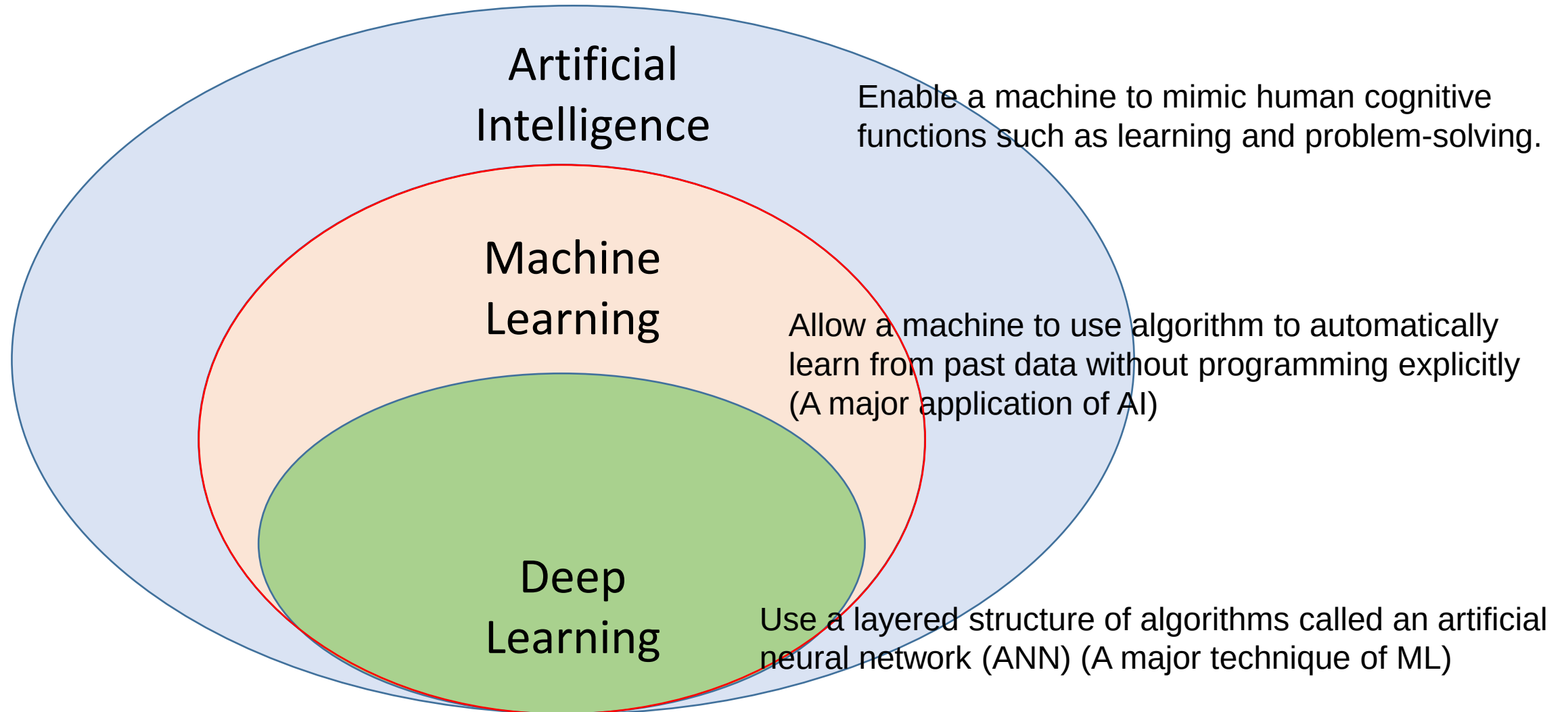
- Support Vector Machine (1995)
- Graphical models
- Bayesian Network
- Topic Modeling (2002)
- Chess-playing case
 - Deep Blue Beats Kasparov (1997)

Deep Learning: Since 2010s

- Two conditions boost AI
 - Big data
 - Computer power
- Deep learning (computer vision, NLP, robotics)
 - 2012, **AlexNet** competed in the [ImageNet Large Scale Visual Recognition Challenge](#)
 - Demonstrate the power of deep learning
 - Accurate training of DNN with GPUs
 - 2014, generative adversarial network (GAN)
 - 2016, **AlphaGo** defeated 18-time world champion [Lee Sedol](#) on Go game
 - 2020, AlphaFold, breakthrough AI solution to a 50-year-old grand challenge in biology
 - 2022, ChatGPT, the fastest-growing consumer software application in history
 - There are more: Midjourney, DALL-E, Stable Diffusion ...

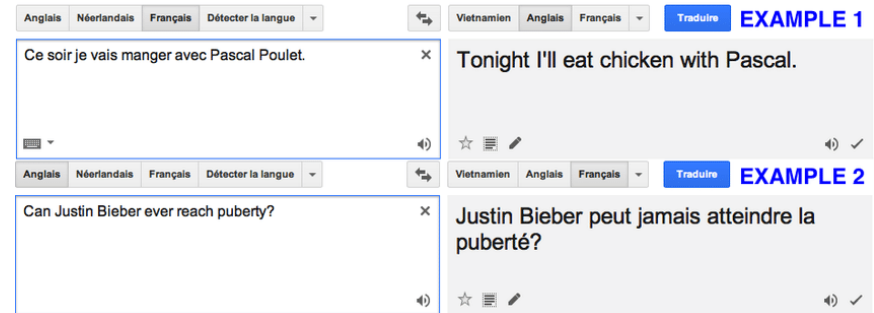


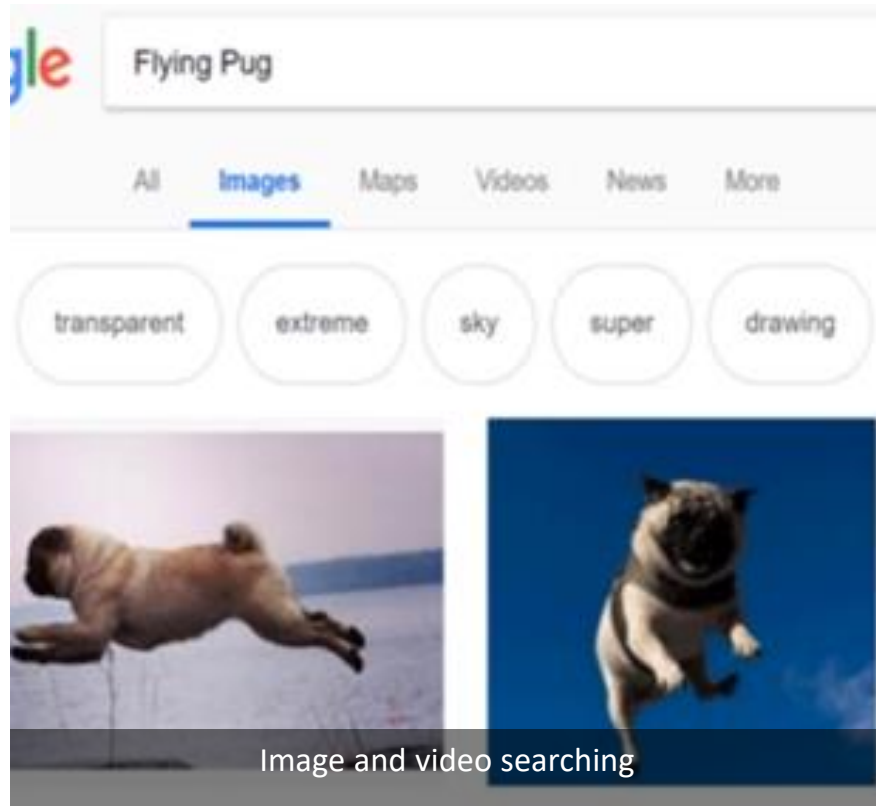
AI vs. ML vs. Deep Learning



Machine Learning is Everywhere

- Speech technologies (e.g. Siri)
 - Automatic speech recognition (ASR)
 - Text-to-speech synthesis (TTS)
 - Dialog systems
- Language processing technologies
 - Question answering
 - Machine translation
- Web search
- Text classification, spam filtering, etc...

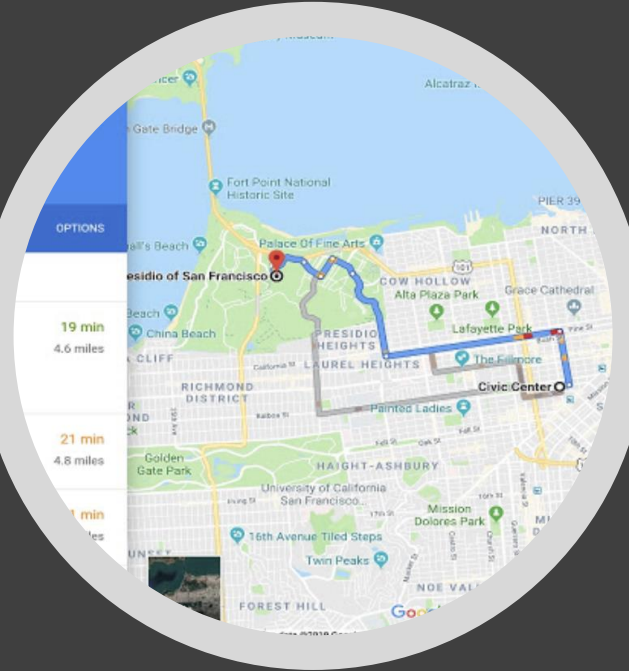




Computer vision



NETFLIX



Berkeley, CA 94709

Tuesday 2:00 PM

Mostly Sunny

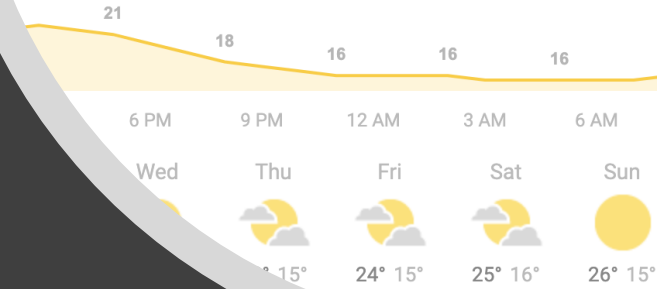
 22°C | °F

Precipitation:

Humidity: 58%

Wind: 24 km/h

Temperature



Tools for Predictions & Decisions

More pervasive than you think

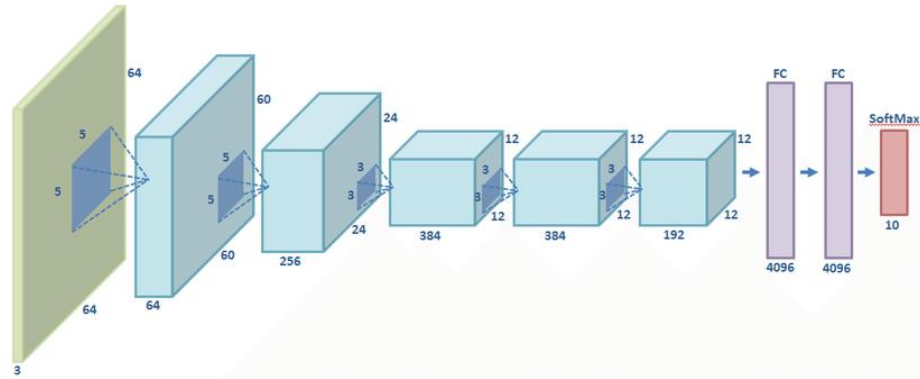
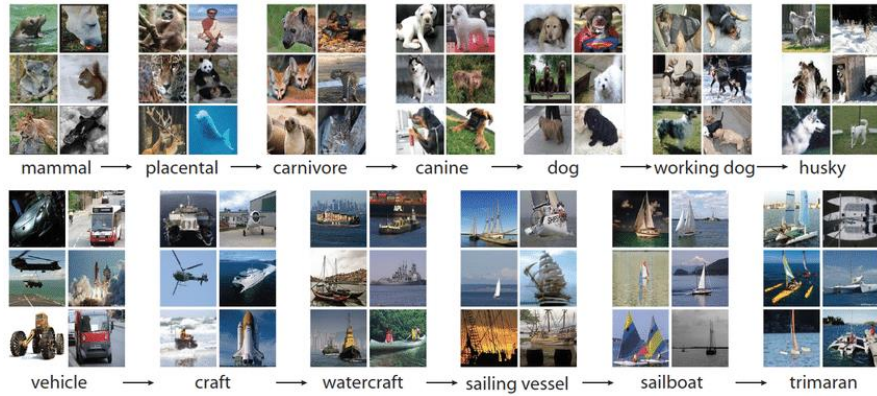


Sent Mail
Spam (372)
Trash

Reasons for Tremendous Advances in ML?

- Big data
 - ImageNet has 14 million images have been hand-annotated
 - Text data available on Internet, eg. Wikipedia
 -
- Machine (deep) learning models
 - AlexNet, Residual Nets, GANs, Attention, BERT
- Computer power
 - GPUs
 - Deep learning frameworks, eg, PyTorch, TensorFlow

AlexNet in ImageNet Challenge (2012)



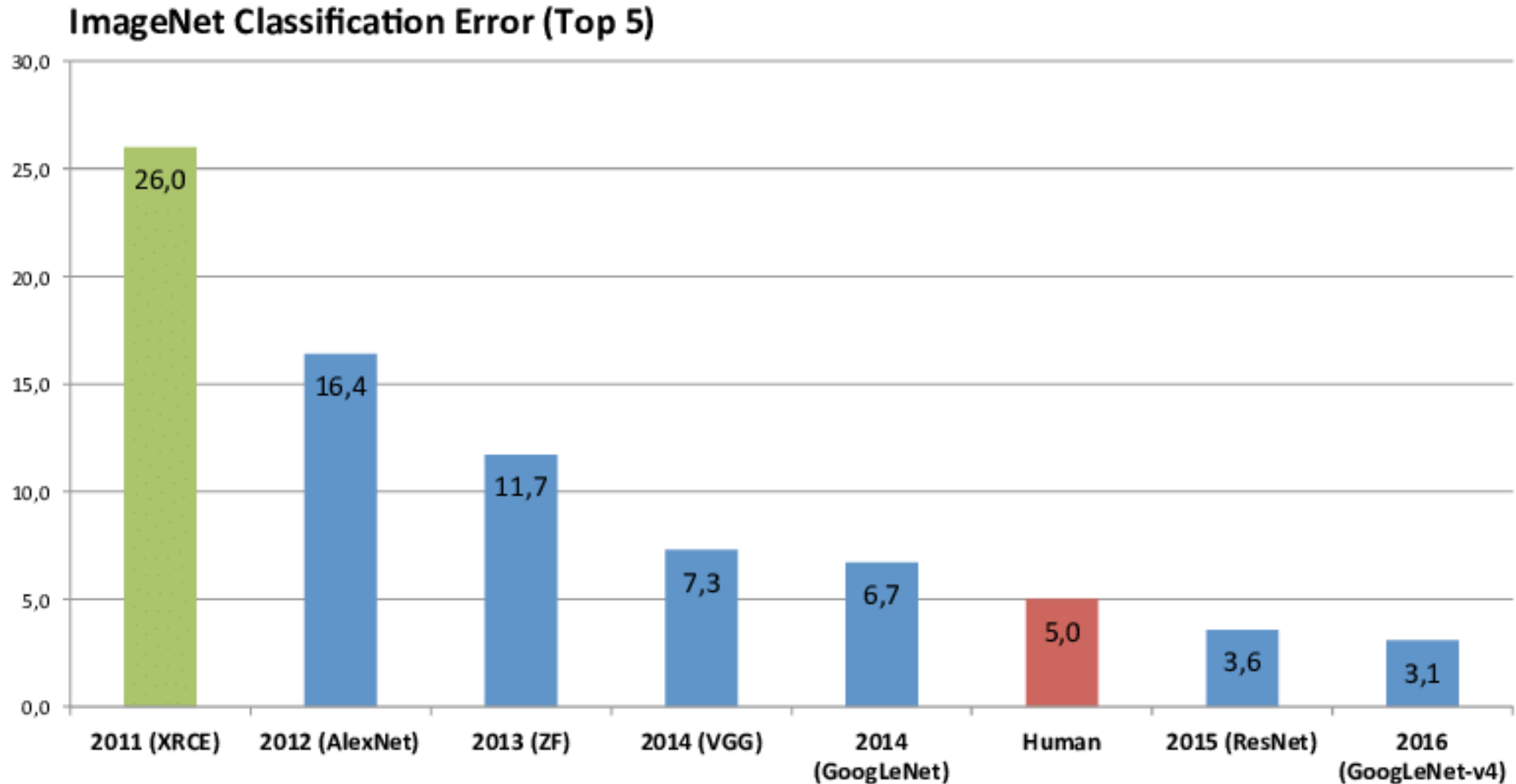
Data: ImageNet, 14 million hand-annotated images

Deep learning models:
AlexNet

Computer power:
GPUs

- Won the competition by a large margin--15.3% VS 26.2% (second place) error rates
- Demonstrate the power of deep learning
- Accurate training of deep neural networks with GPUs

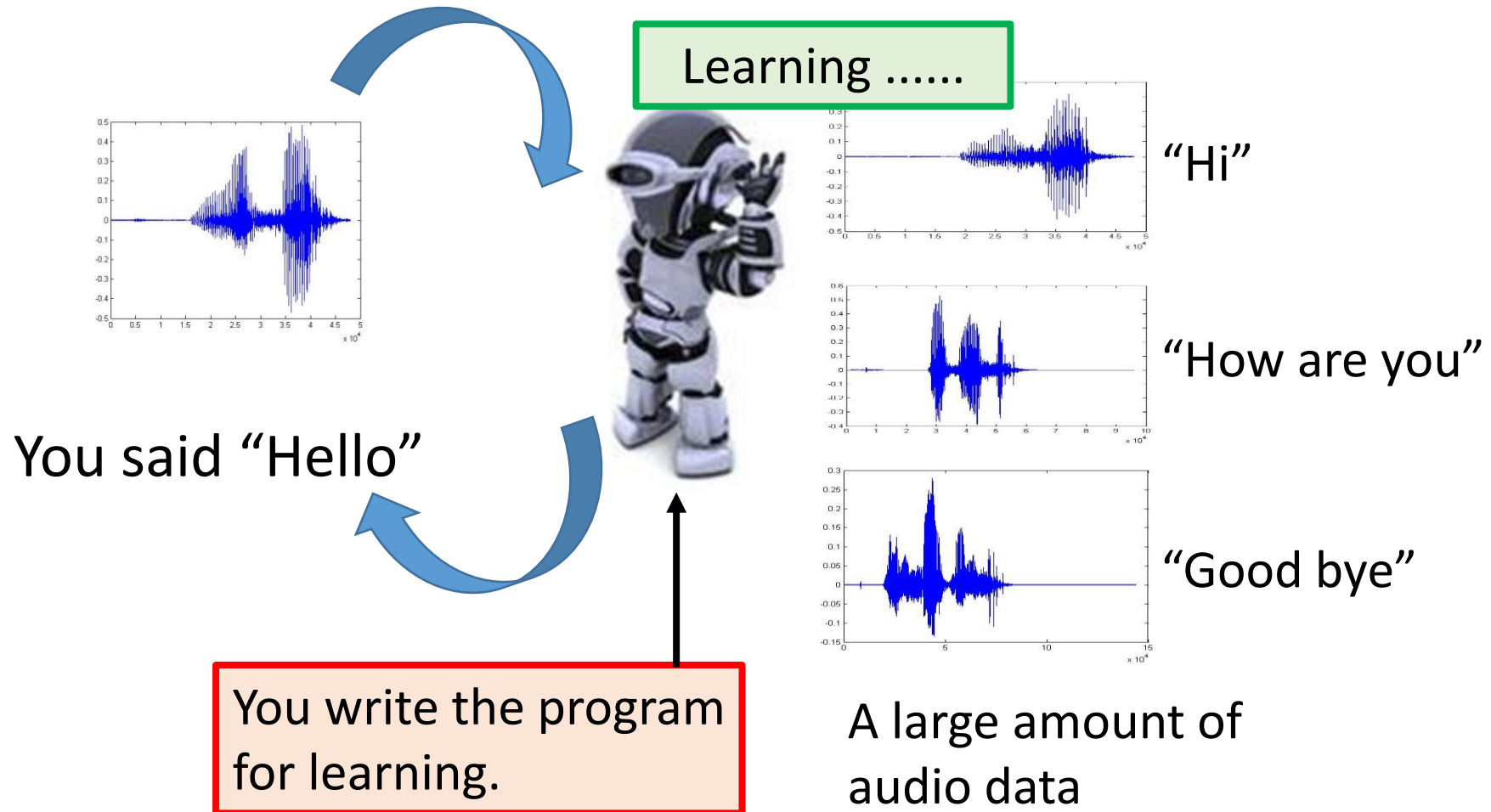
Progress in Image Recognition



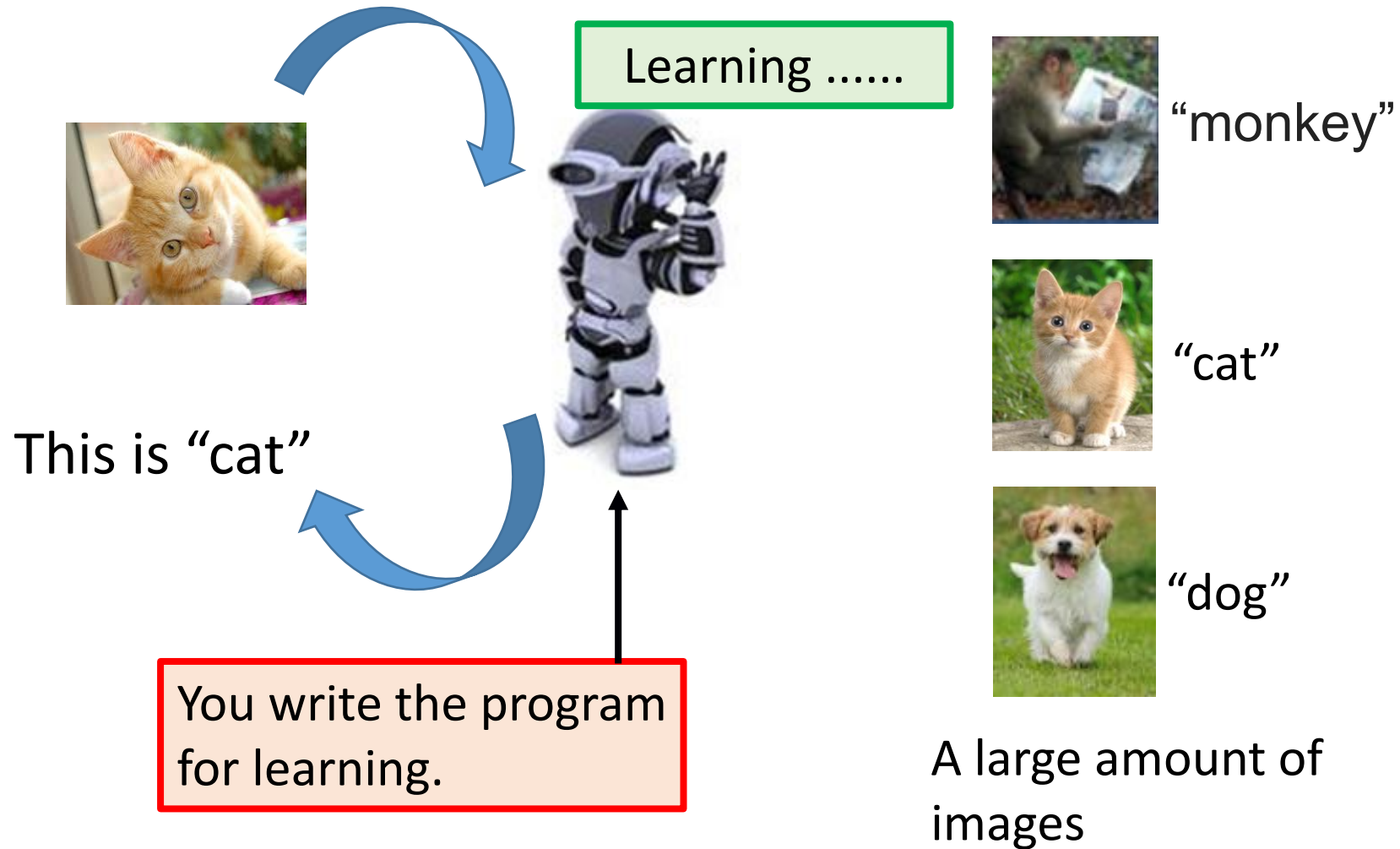
Machine Learning

- Machine learning aims to build a mathematical (statistical) model based on sample **data**, known as "training data", to make predictions or decisions
- Machine learning is the process that powers many of the services we use today
 - Recommendation systems like those on Netflix, YouTube, and Spotify;
 - Search engines like Google;
 - Social-media feeds like Facebook and Twitter;
 - Voice assistants like Siri and Alexa

Machine Learning: Speech Recognition Example

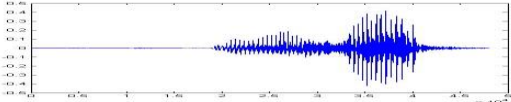


Machine Learning : Image Recognition Example



Machine Learning $\approx$ Look for Function

- Speech Recognition

$$f(\text{  }) = \text{"How are you"}$$

- Image Recognition

$$f(\text{  }) = \text{"Cat"}$$

- Playing Go

$$f(\text{  }) = \text{"5-5"}_{\text{(next move)}}$$

- Dialogue System

$$f(\text{ "How are you?" }_{\text{(what the user said)}}) = \text{"I am fine."}_{\text{(system response)}}$$

Start with a Project

Image Recognition:

$$f\left(\text{img of cat}\right) = \text{"cat"}$$

A set of
function

Model

$f_1, f_2 \dots$ Different parameters

f_1



) = "cat"

f_2



) = "monkey"

f_1



) = "dog"

f_2

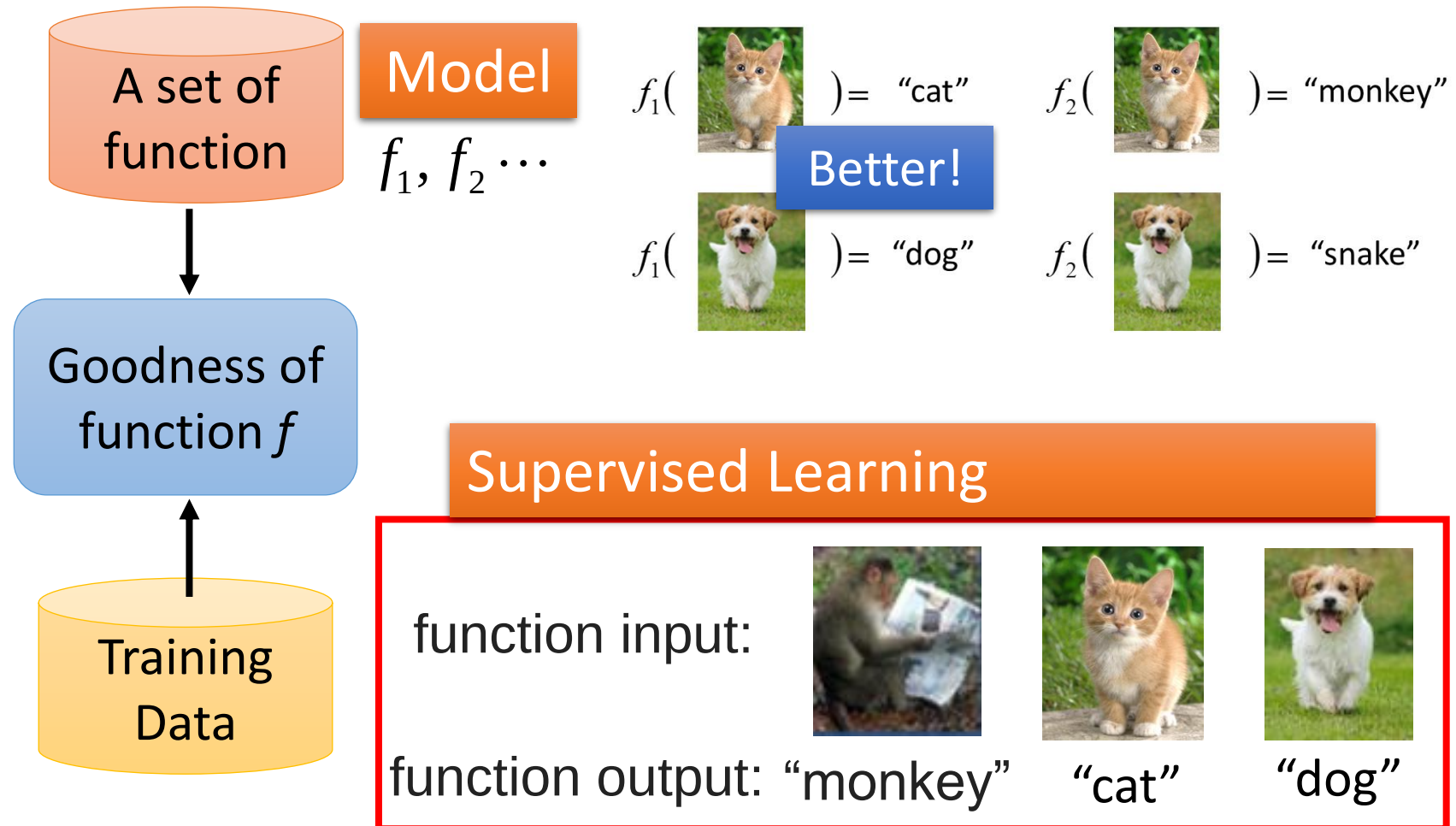


) = "snake"

Framework

Image Recognition:

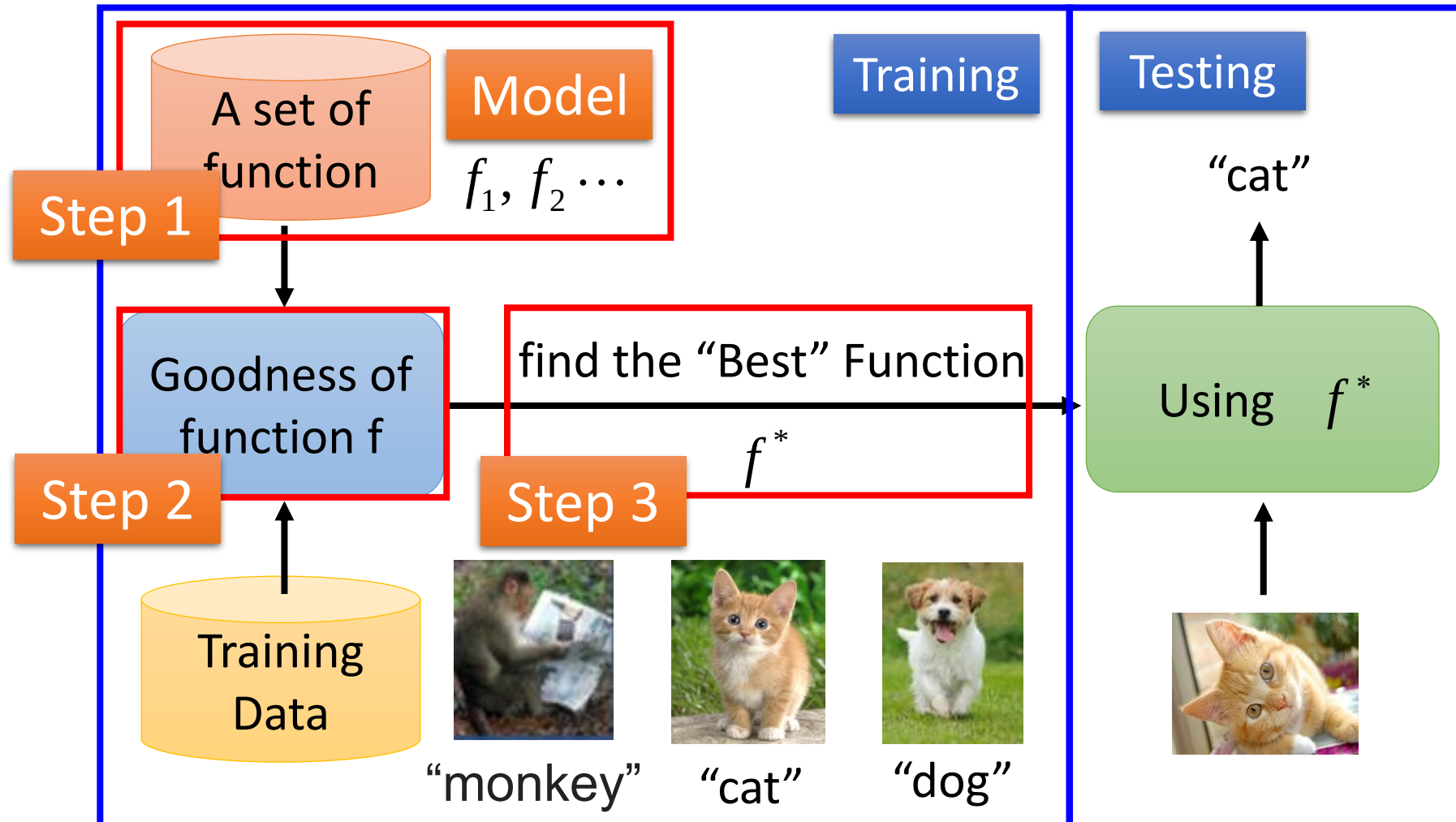
$$f\left(\text{img_cat}\right) = \text{"cat"}$$



Framework

Image Recognition:

$$f\left(\text{Image of a cat}\right) = \text{"cat"}$$



Step 0 - Problem Formulation

Step 0: What kind of function do you want to find?

Application oriented

- Different data
- Different Tasks

Step 1:
define a set
of function

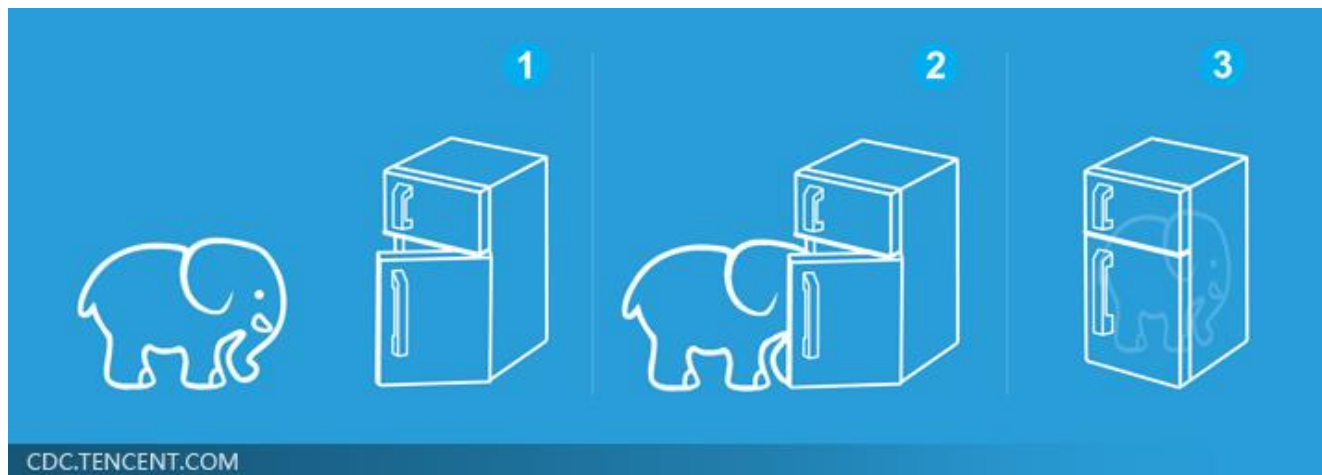


Step 2:
goodness of
function



Step 3: pick
the best
function

Just like the three steps to put an elephant into the fridge.....



(Big) Data is Everywhere...



processed about over 20
petabytes of data per day

Twitter now sends and
receives as many as 500
million “tweets” every day.



As of January 2013, Facebook users had
uploaded over 240 billion photos, with 350
million new photos every day.



S3: 449B objects, peak 290k
request/second (7/2011)
1T objects (6/2012)



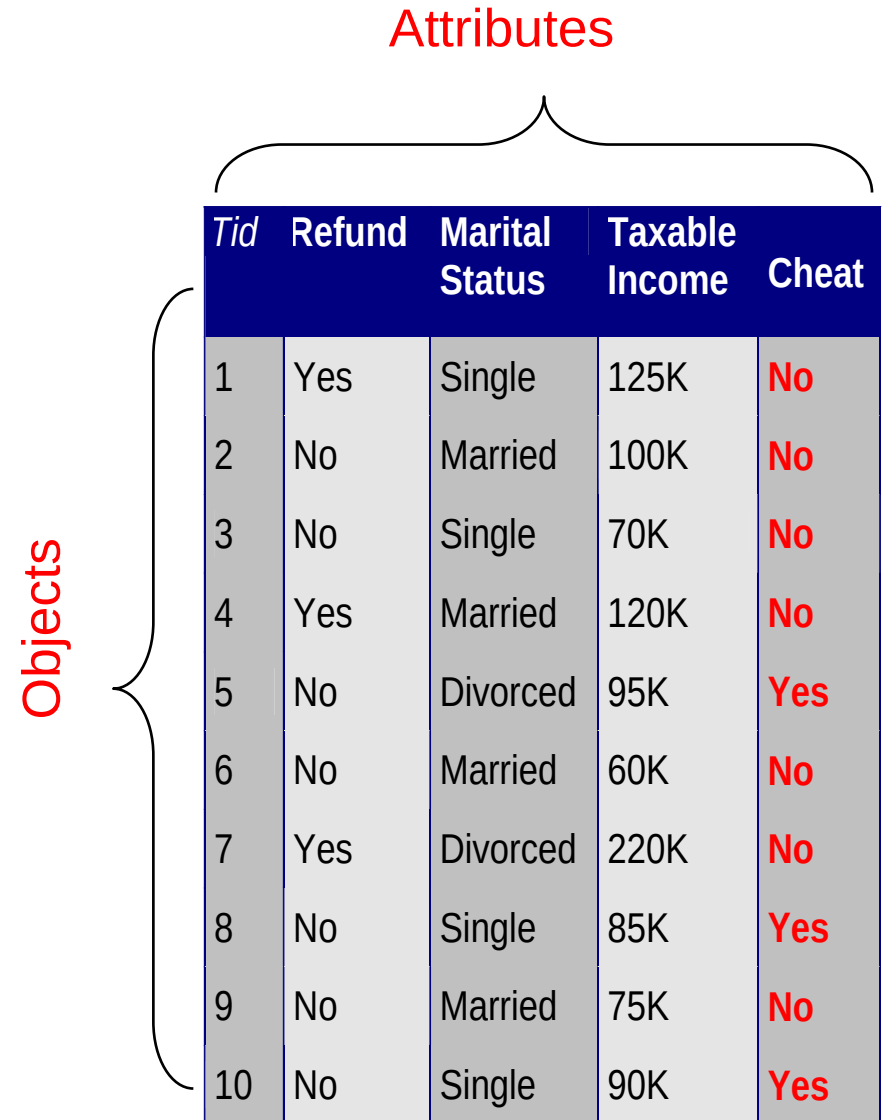
transfers about 30 petabytes of
data through its networks each
day.

150 PB on 50k+ servers
running 15k apps (6/2011)



What is Data?

- Collection of data **objects** and their **attributes**
- An attribute is a property or characteristic of an object
 - Examples: eye color of a person, temperature, etc.
 - Attribute is also known as variable, field, characteristic, or feature
- A collection of attributes describe an object
 - Object is also known as record, point, case, sample, entity, or instance



Attributes				
Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Attributes

- **Attribute (or dimensions, features, variables):**
 - a data field, representing a characteristic or feature of a data object.
 - *E.g., customer_ID, name, address*
- **Types:**
 - Nominal
 - Binary
 - Ordinal
 - Numeric

Attribute Values

- Attribute values are **numbers** or **symbols** assigned to an attribute for a particular object
- Distinction between *attributes* and *attribute values*
 - Same attribute can be mapped to different attribute values
 - Example: height can be measured in feet or meters
 - Different attributes can be mapped to the same set of values
 - Example: Attribute values for ID and age are integers
 - But properties of attribute values can be different

Attribute Types

- **Nominal:** used for naming or labelling variables, without any quantitative value
 - Categories, states, or “names of things”
 - *Hair_color* = {auburn, black, blond, brown, grey, red, white}
 - marital status, occupation, ID numbers, zip codes
- **Binary**
 - Nominal attribute with only 2 states (0 and 1)
 - e.g., medical test (positive vs. negative)
- **Ordinal**
 - Values have a meaningful order (ranking) but magnitude between successive values is not known.
 - *Size* = {small, medium, large}, grades = {A, B, C, D, F}
- **Numeric**
 - Continuous: real numbers such as speed
 - Discrete: can only take certain values

Types of Data Sets

- **Record**

- Relational records
- Data matrix, e.g., numerical matrix, crosstabs
- Document data: text documents: term-frequency vector
- Transaction data

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

- **Graph and network**

- World Wide Web
- Social or information networks
- Molecular Structures



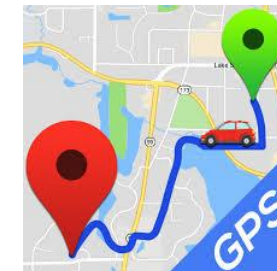
- **Ordered**

- Video data: sequence of images
- Temporal data: time-series
- Sequential Data: transaction sequences
- Genetic sequence data



- **Spatial, image and multimedia:**

- Spatial data: maps
- Image data:
- Video data:



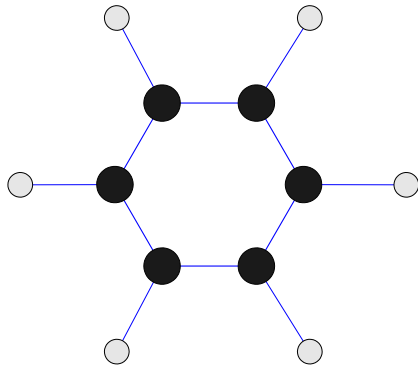
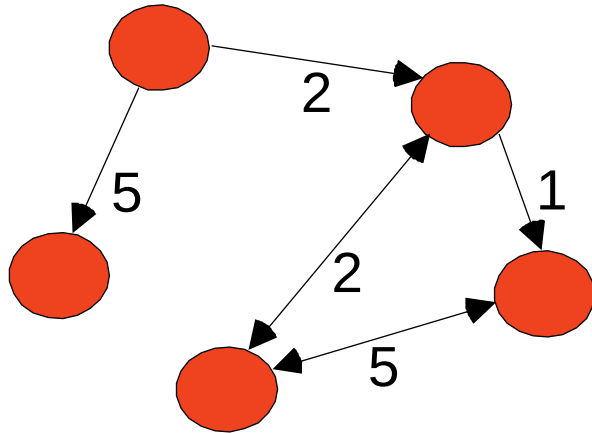
Record Data

- Data that consists of a collection of records, each of which consists of a fixed set of attributes

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Graph Data

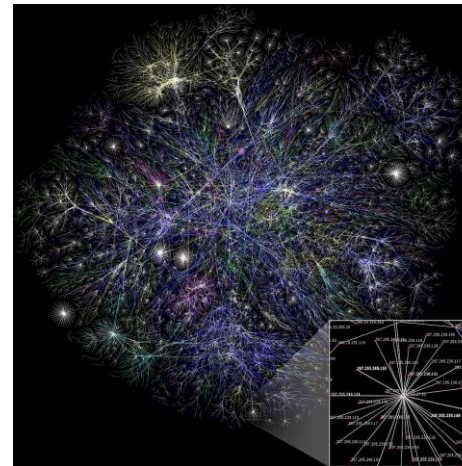
- Examples: Social networks, Generic graph, a Molecule, and Webpages



Benzene Molecule: C₆H₆



Social networks



part of the Internet

Ordered (Sequential) Data



time
series

“Natural language processing is a subfield of linguistics, computer science, and artificial intelligence concerned with the interactions between computers and human language, in particular how to program computers to process and analyze large amounts of natural language data.”

human
language

Spatial Data



map



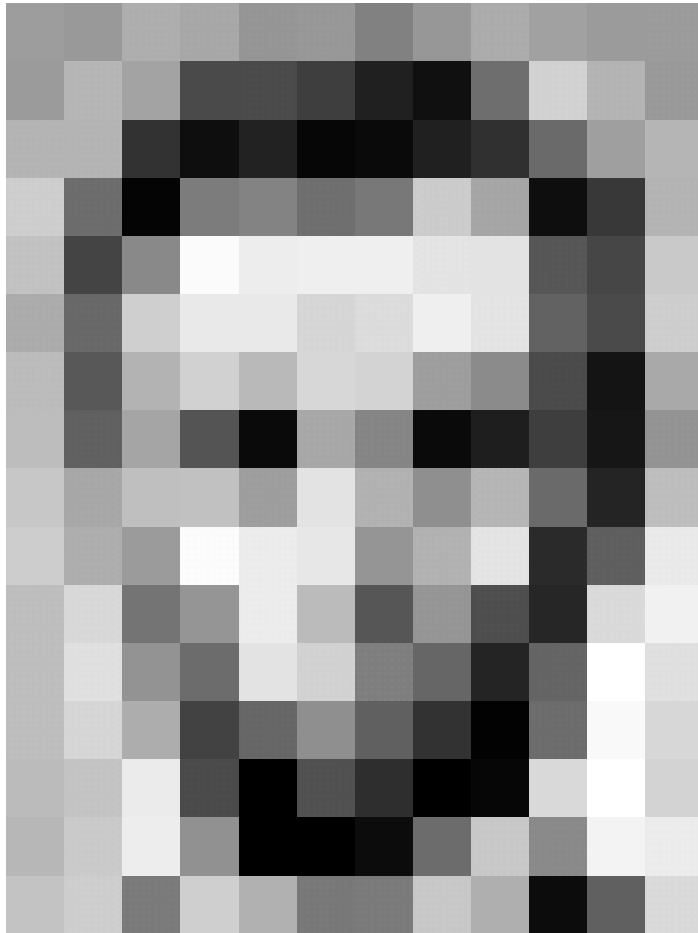
images

Data Matrix

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute
- Such data set can be represented by an $m \times n$ matrix, where there are **m rows**, one for each object, and **n columns**, one for each attribute

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

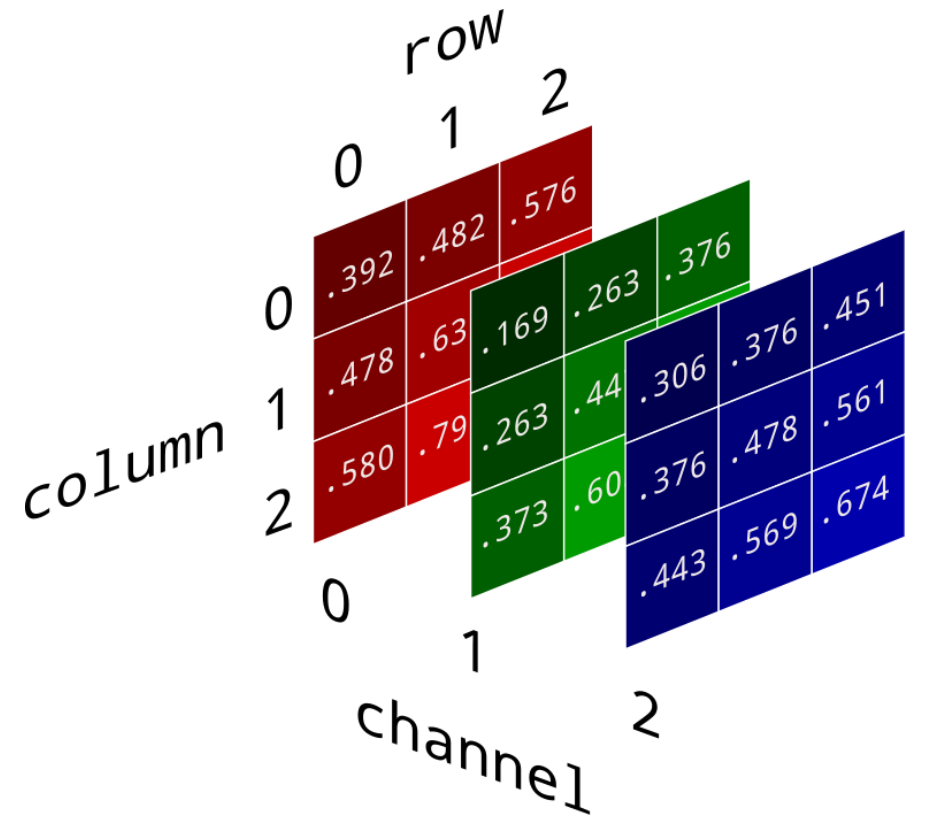
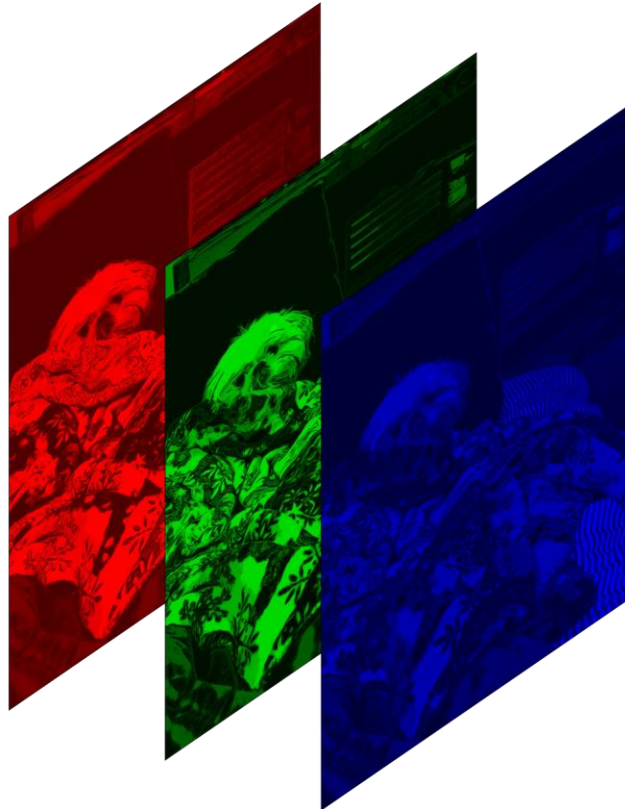
Image Data



157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	205	123	207	177	121	123	200	175	13	96	218

157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	205	123	207	177	121	123	200	175	13	96	218

Image Data: Color Images



Document Data

- Each document becomes a `term' vector,
 - each term is a component (attribute) of the vector,
 - the value of each component is the number of times the corresponding term occurs in the document.

D1 = "I really, really, like math"

D2 = "I hate math",

	I	really	like	hate	math
D1	1	2	1	0	1
D2	1	0	0	1	1

Step 0 - Problem Formulation

Step 0: What kind of function do you want to find?

Application oriented

- Different data
- Different Tasks

Step 1:
define a
function

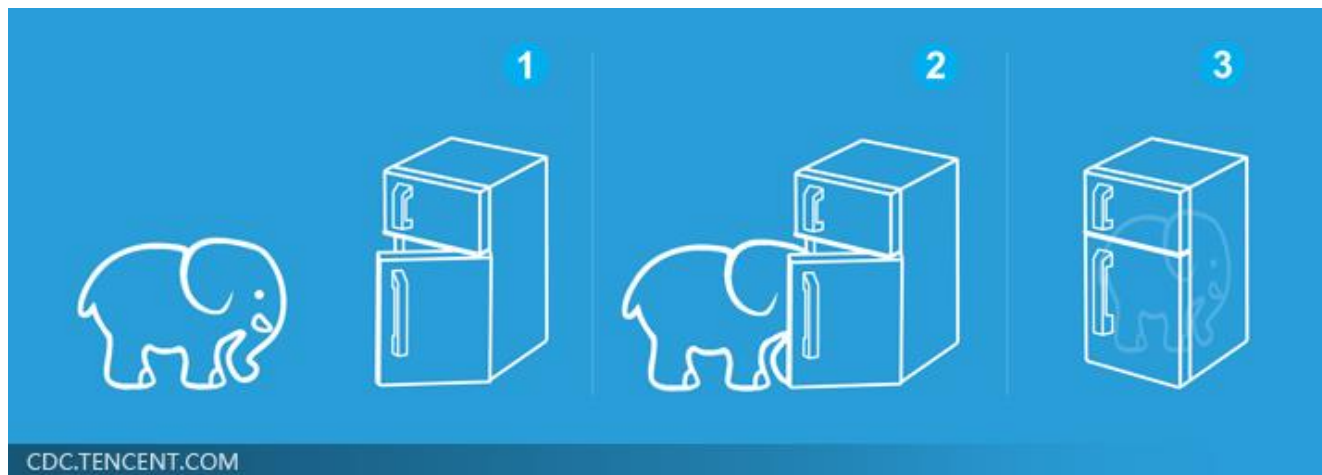


Step 2:
goodness of
function



Step 3: find
the best
function

Just like the three steps to put an elephant into the fridge.....



Task: Regression

Regression

The output of the target function f is “scalar”.

Predict
PM2.5



Training Data:

Input:

9/01 PM2.5 = 63 9/02 PM2.5 = 65

Input:

9/12 PM2.5 = 30 9/13 PM2.5 = 25

Output:

9/03 PM2.5 = 100

Output:

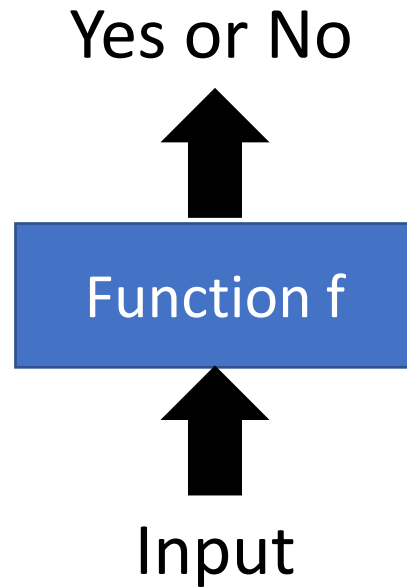
9/14 PM2.5 = 20

Task: Classification

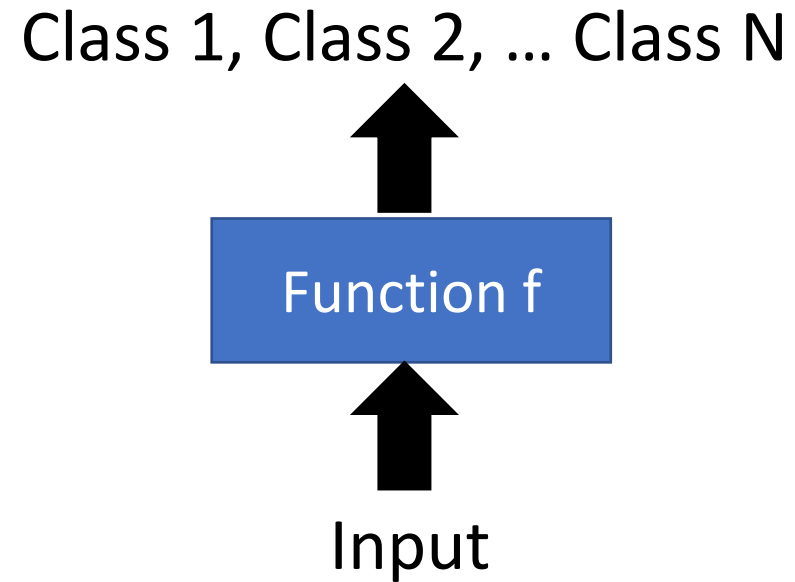
- Given a collection of records (training set)
 - Each record is characterized by a tuple (x, y) , where x is the attribute set and y is the class label
 - x : attribute, predictor, independent variable, input
 - y : class, response, dependent variable, output
- Task:
 - Learn a model that maps each attribute set x into one of the predefined class labels y

Task: Classification

- Binary Classification

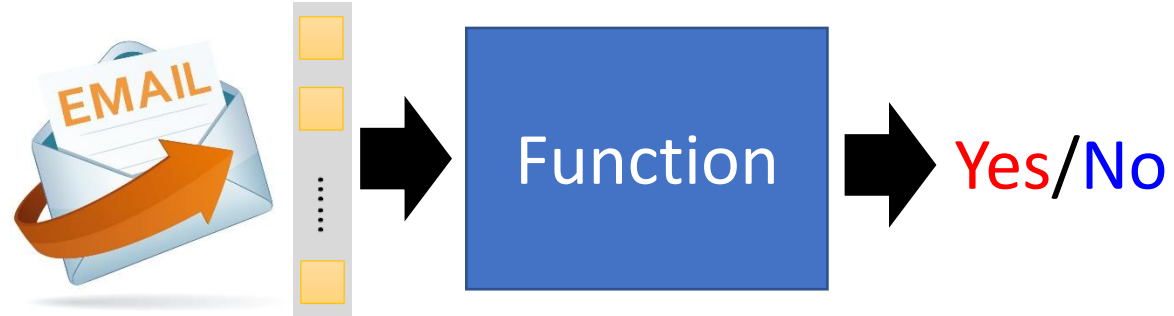


- Multi-class Classification



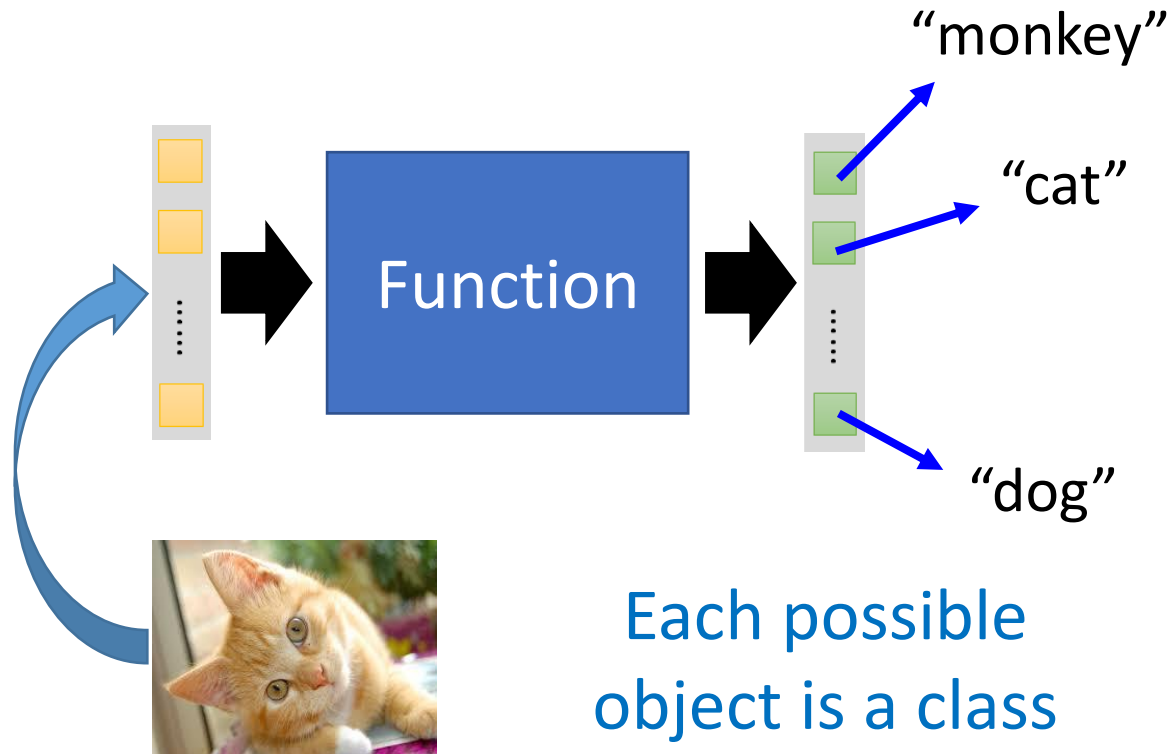
Binary Classification

Spam
filtering

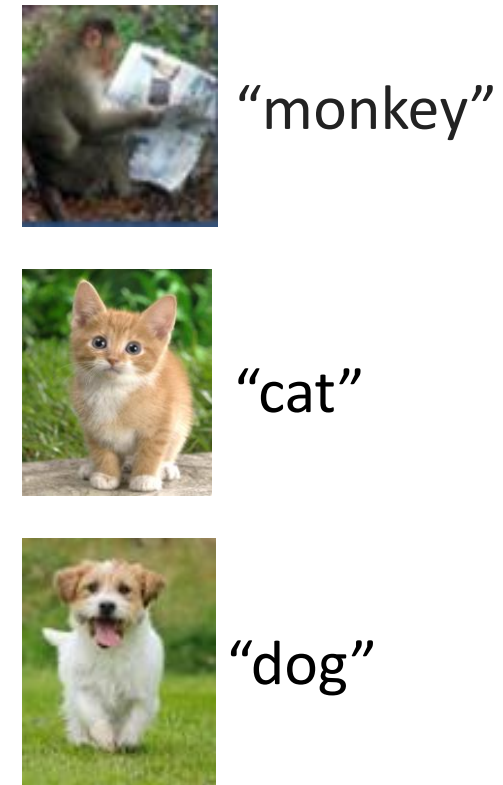


Multi-Class Classification

Image Recognition



Training Data



Multi-Class Classification

Go Play



Playing GO



Each position
is a class
(19 x 19 classes)



a position on
the board

Next move

Step 1 – Function with Unknow Parameters

Step 0: What kind of function do you want to find?

Application oriented

- Different data
- Different Tasks

Step 1:
define a
function

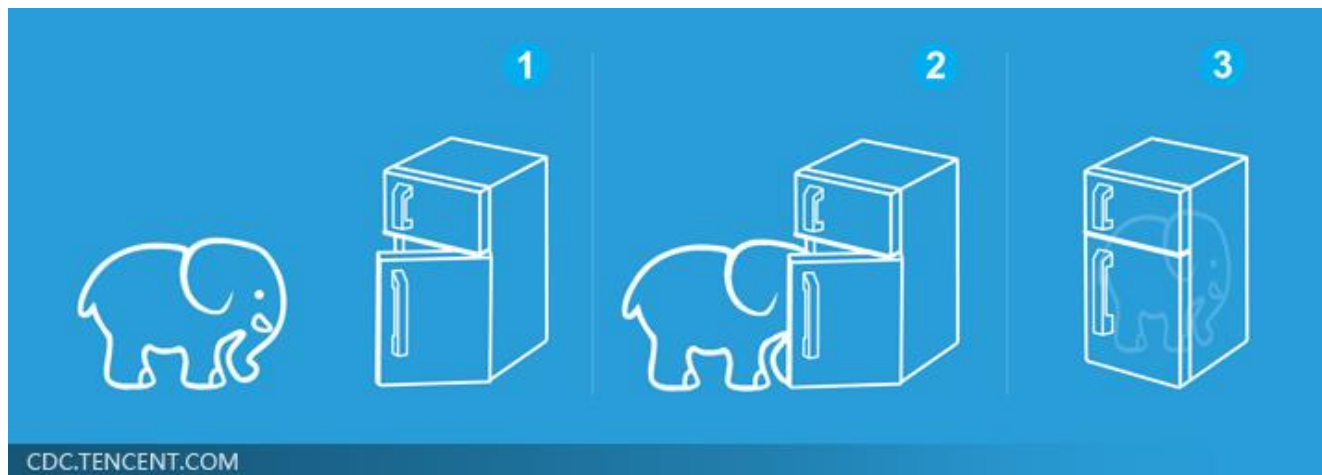


Step 2:
goodness of
function



Step 3: find
the best
function

Just like the three steps to put an elephant into the fridge.....



The function we want to find ...

$y = f($
Return on
Monday?

Tesla, Inc. (TSLA) ☆ Follow ↕ Compare 🕒 Time to buy TSLA?						
412.38 -2.73 (-0.66%) 411.91 -0.47 (-0.11%)						
At close: January 23 at 4:00:00 PM EST After hours: January 23 at 7:59:58 PM EST ⌵						
Jan 24, 2024 - Jan 24, 2025 ▾ Historical Prices ▾ Daily ▾						
Currency in USD 📄 Download						
Date	Open	High	Low	Close ⓘ	Adj Close ⓘ	Volume
Jan 23, 2025	416.06	420.73	408.95	412.38	412.38	50,554,400
Jan 22, 2025	416.81	428.00	414.59	415.11	415.11	60,963,300
Jan 21, 2025	432.64	433.20	406.31	424.07	424.07	87,320,900
Jan 17, 2025	421.50	439.74	419.75	426.50	426.50	94,991,400
Jan 16, 2025	423.49	424.00	409.13	413.82	413.82	68,335,200
Jan 15, 2025	409.90	429.80	405.66	428.22	428.22	81,375,500
Jan 14, 2025	414.34	422.64	394.54	396.36	396.36	84,565,000
Jan 13, 2025	383.21	403.79	380.07	403.31	403.31	67,580,500
Jan 10, 2025	391.40	399.28	377.29	394.74	394.74	62,287,300
Jan 8, 2025	392.95	402.50	387.40	394.94	394.94	73,038,800
Jan 7, 2025	405.83	414.33	390.00	394.36	394.36	75,699,500
Jan 6, 2025	423.20	426.43	401.70	411.05	411.05	85,516,500
Jan 3, 2025	381.48	411.88	379.45	410.44	410.44	95,423,300

Function with Unknown Parameters

$$y = f($$



Model $y = b + wx_1$ based on domain knowledge

feature

y : close price on Jan 27, x_1 : close price on Jan 24

w and b are unknown parameters (learned from data)

weight **bias**

Tesla, Inc. (TSLA) ☆ Follow ↔ Compare Time to buy TSLA?						
412.38 -2.73 (-0.66%) 411.91 -0.47 (-0.11%)						
At close: January 23 at 4:00:00 PM EST After hours: January 23 at 7:59:58 PM EST						
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Jan 7, 2025	405.83	414.33	390.00	394.36	394.36	75,699,500
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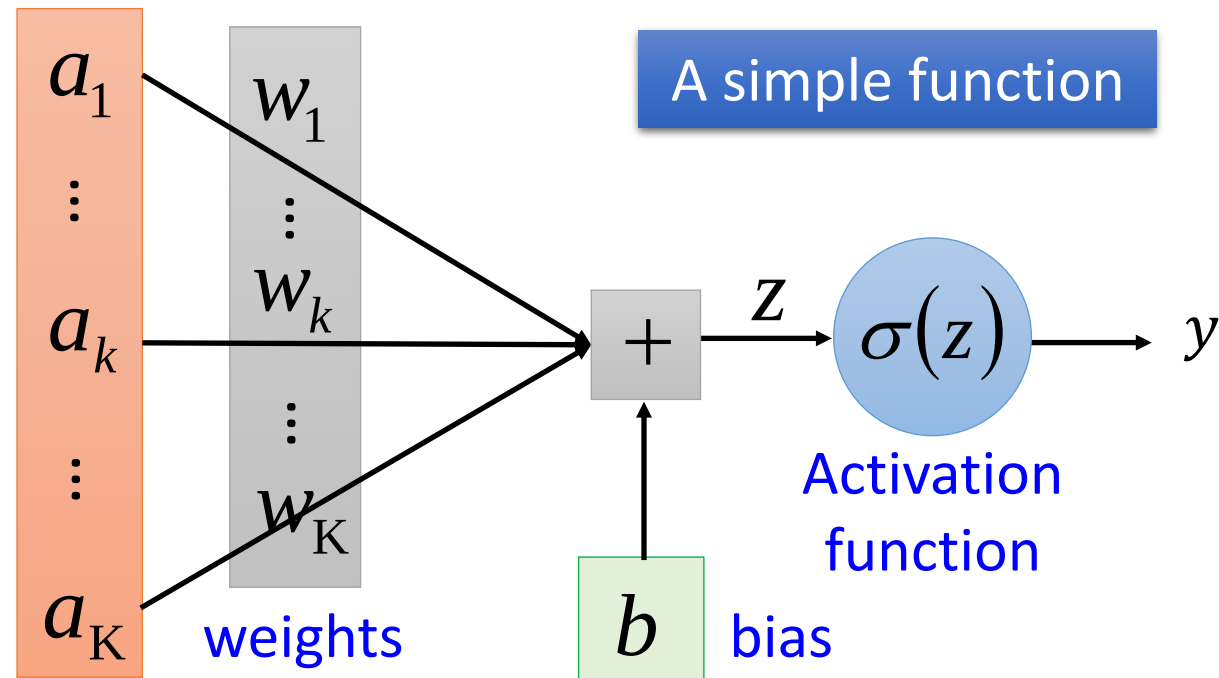
Classification: Handwritten Digit Recognition

4 → 4	2 → 2	3 → 3
4 → 4	9 → 9	0 → 0
5 → 5	7 → 7	1 → 1
9 → 9	0 → 0	3 → 3
6 → 6	7 → 7	4 → 4

Neural Network

Neuron

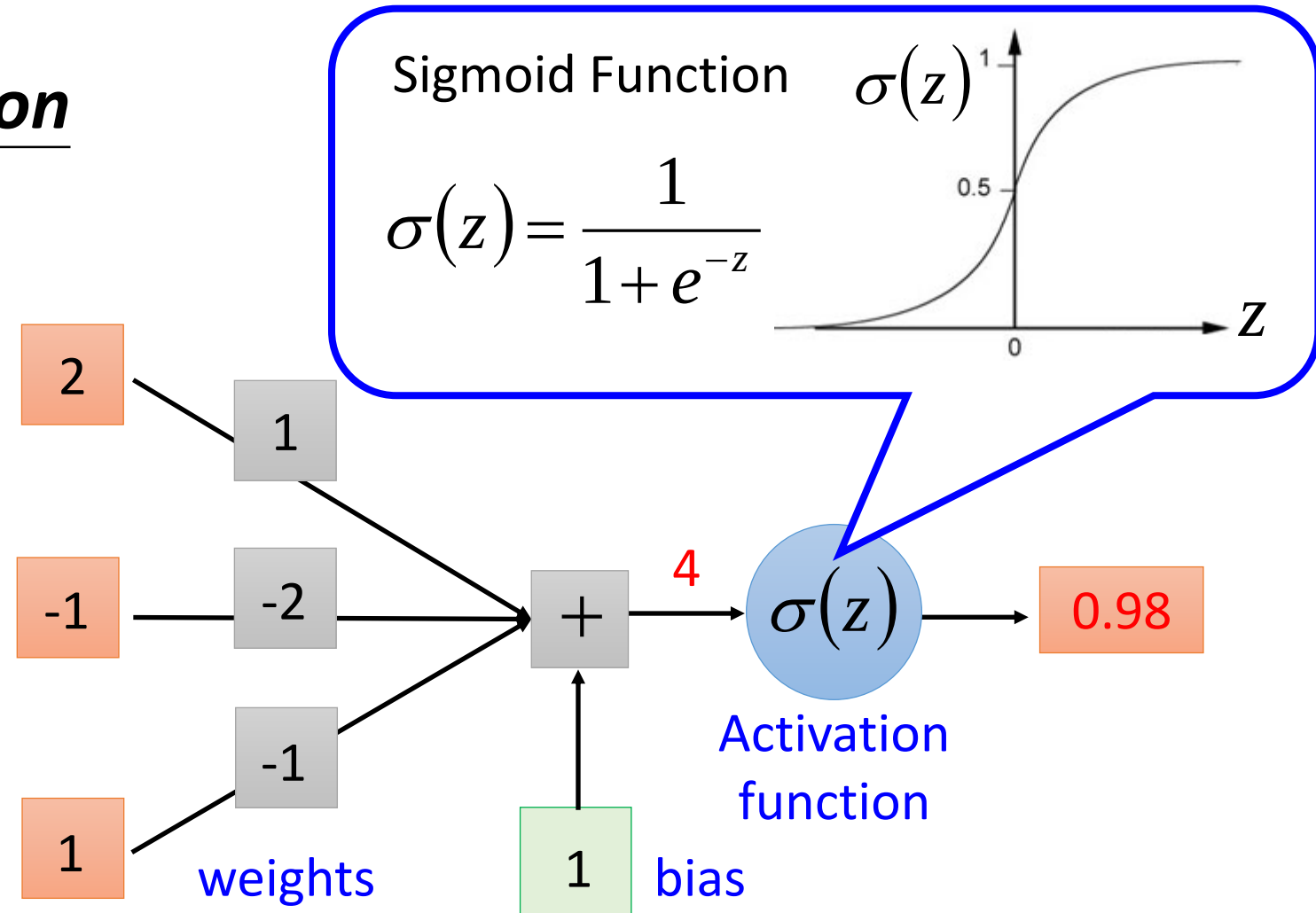
$$z = a_1 w_1 + \dots + a_k w_k + \dots + a_K w_K + b$$



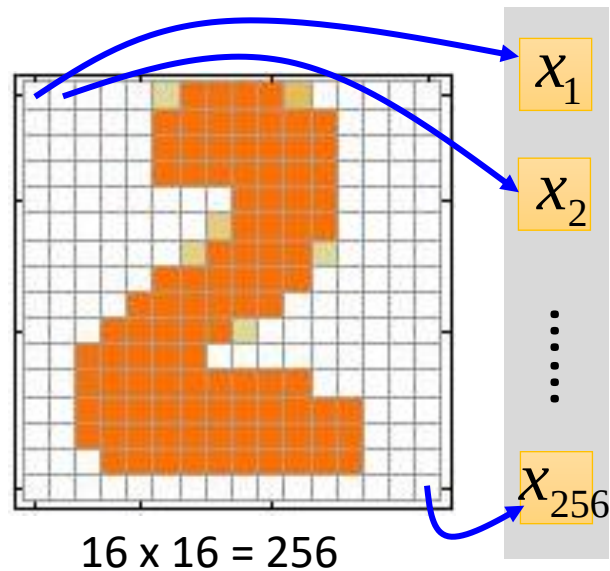
Weights and biases are called network parameters

Neural Network

Neuron

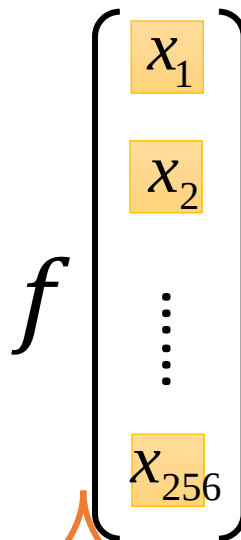


Input

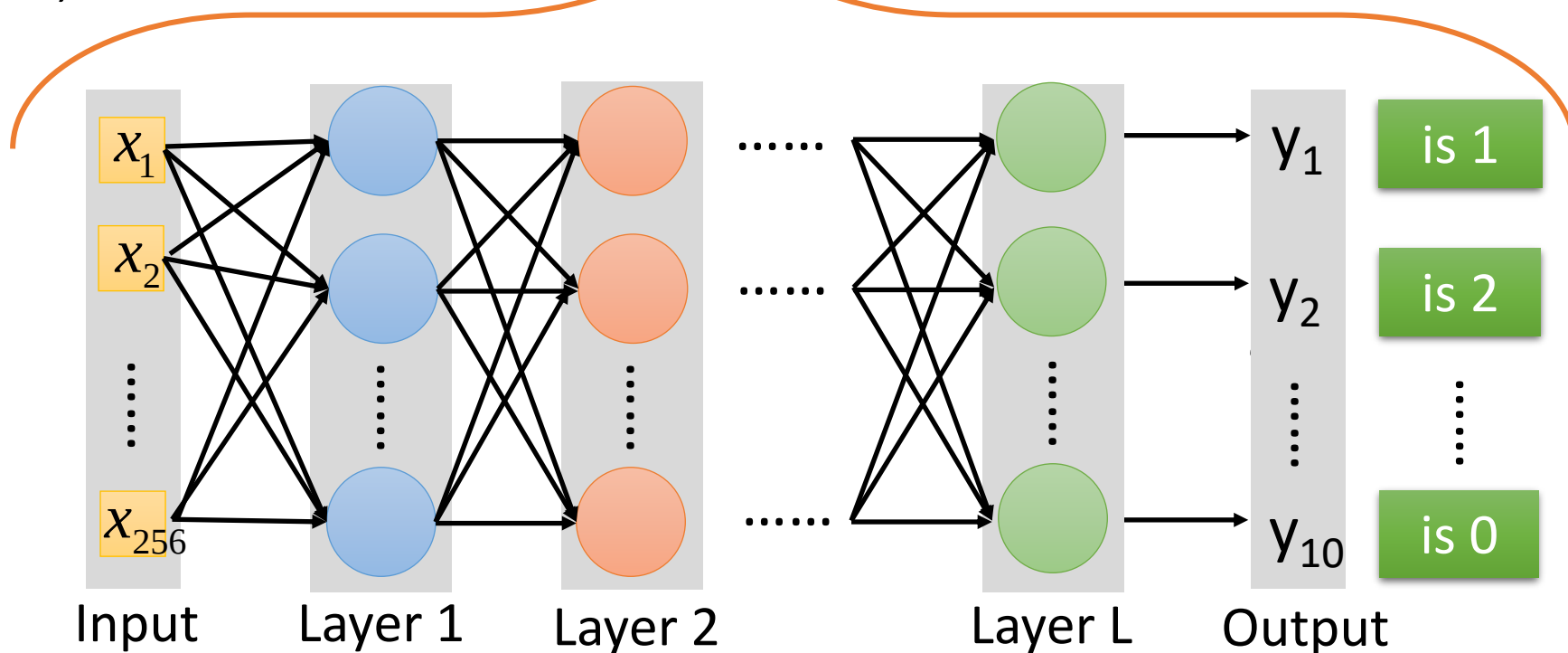
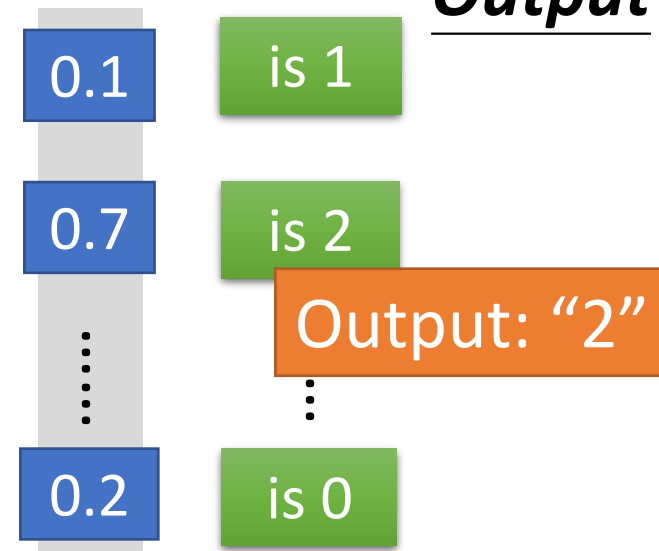


Ink $\rightarrow$ 1, No ink $\rightarrow$ 0

Model



Output



Step 2 - Measure Error

Step 0: What kind of function do you want to find?

Application oriented

- Different data
- Different Tasks

Step 1:
define a
function

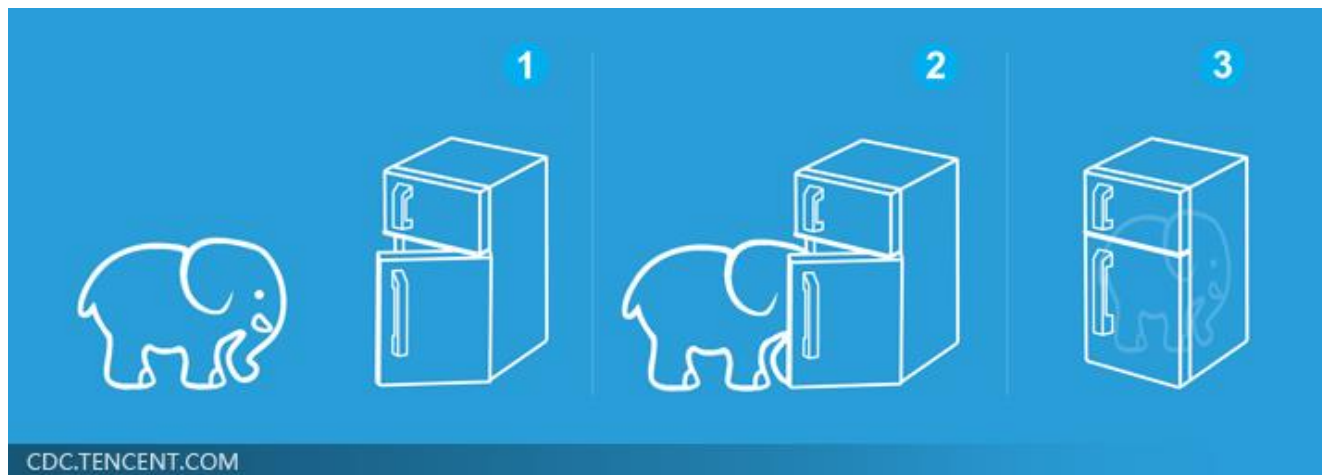


Step 2:
goodness of
function



Step 3: find
the best
function

Just like the three steps to put an elephant into the fridge.....



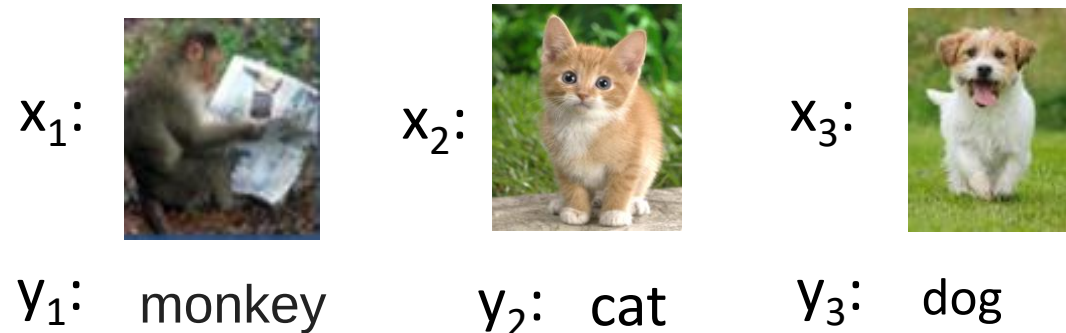
Supervised Learning



Supervised



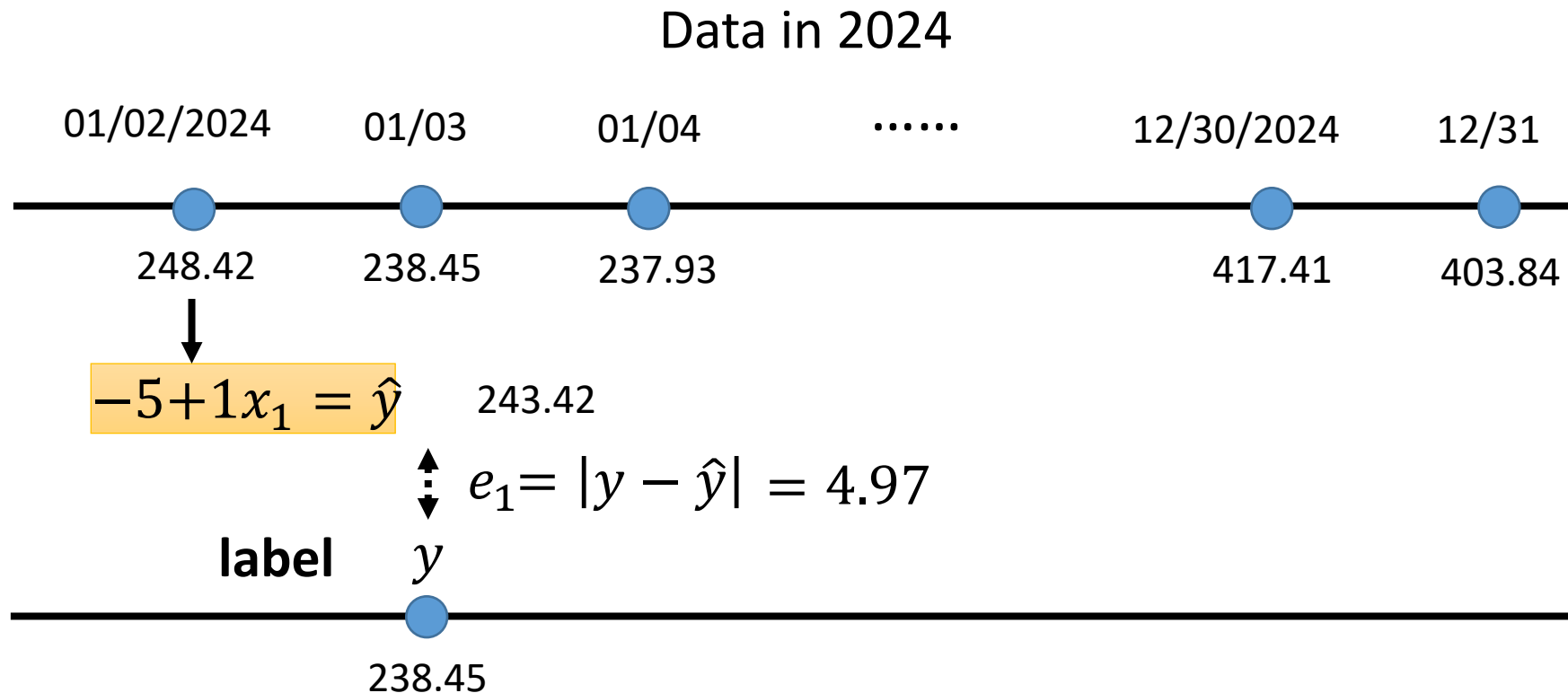
Supervised



Define Loss from Training Data

- Loss is a function of parameters $L(b, w)$
- Loss: how good a set of values is.

$$L(-5, 1) \quad y = b + wx_1 \longrightarrow y = -10 + 1x_1 \quad \text{How good it is?}$$

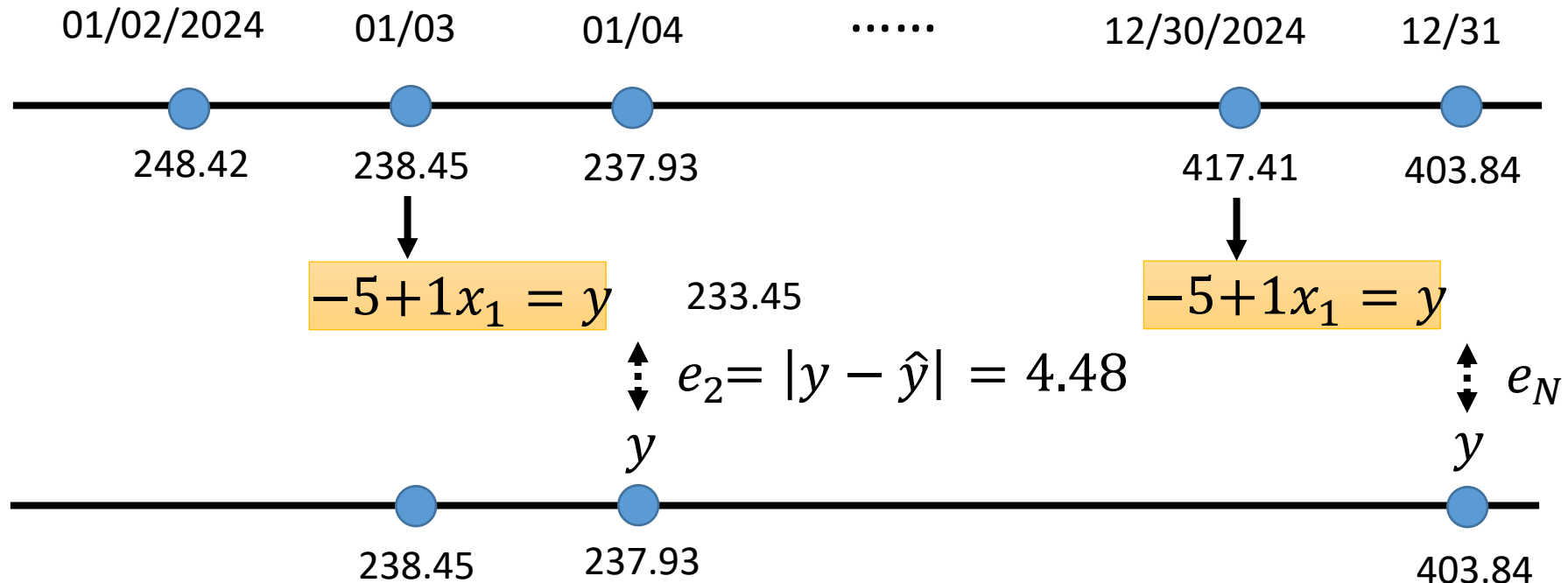


Define Loss from Training Data

- Loss is a function of parameters $L(b, w)$
- Loss: how good a set of values is.

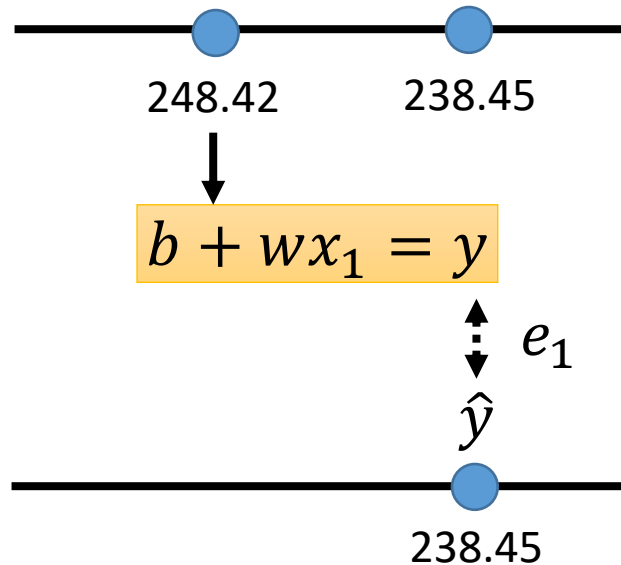
$$L(-5, 1) \quad y = b + wx_1 \longrightarrow y = -10 + 1x_1 \quad \text{How good it is?}$$

Data in 2024



Define Loss from Training Data

- Loss is a function of parameters $L(b, w)$
- Loss: how good a set of values is.



Loss:
$$L = \frac{1}{N} \sum_n e_n$$

$e = |y - \hat{y}|$ L is mean absolute error (MAE)

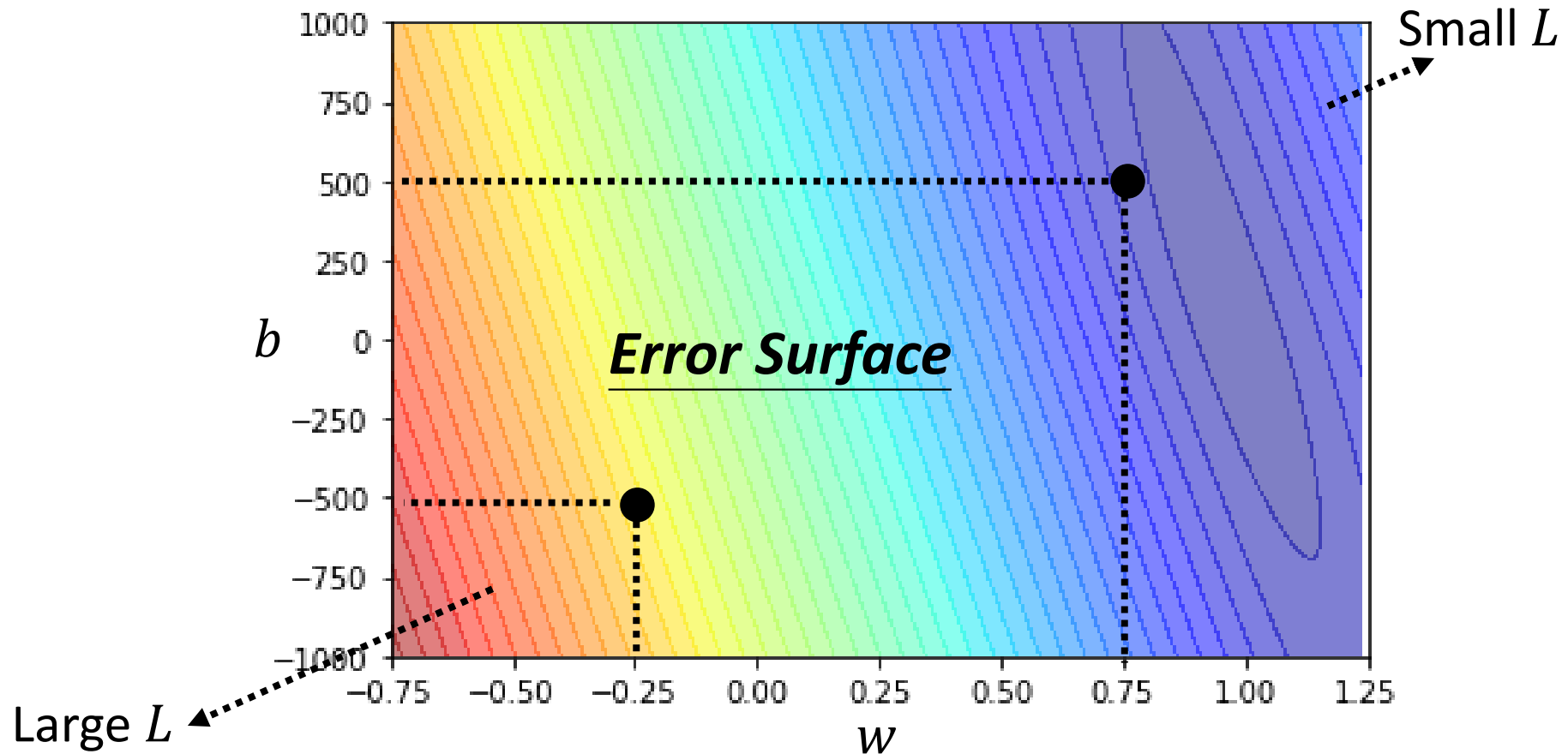
$e = (y - \hat{y})^2$ L is mean square error (MSE)

If y and $\hat{y}$ are both probability distributions $\longrightarrow$ Cross-entropy

Define Loss from Training Data

- Loss is a function of parameters $L(b, w)$
- Loss: how good a set of values is.

Model $y = b + wx_1$



Step 3 - Optimization

Step 0: What kind of function do you want to find?

Step 1:
define a set
of function



Step 2:
goodness of
function

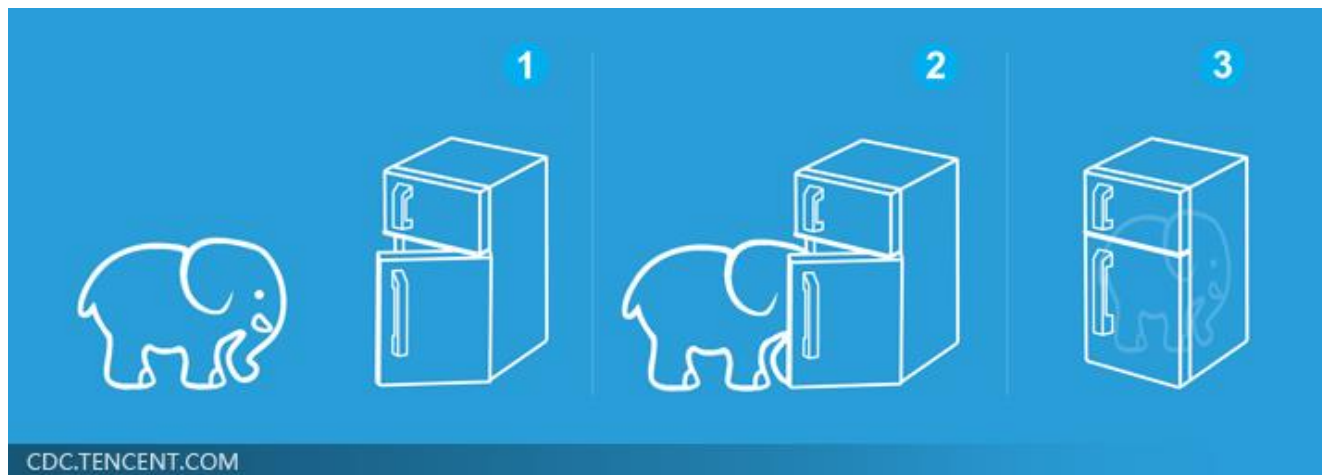


Step 3: find
the best
function

Application oriented

- Different data
- Different Tasks

Just like the three steps to put an elephant into the fridge.....



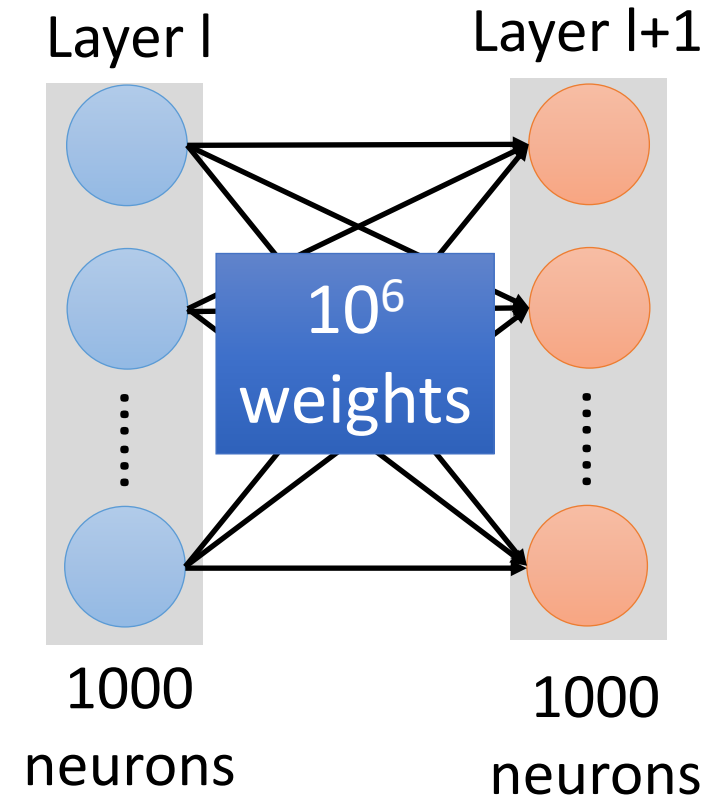
Find the Best Function through Optimization

Enumerate all possible values

Network parameters $\theta =$
 $\{w_1, w_2, w_3, \dots, b_1, b_2, b_3, \dots\}$

Millions of parameters

Today a network can have
more than 100B parameters.

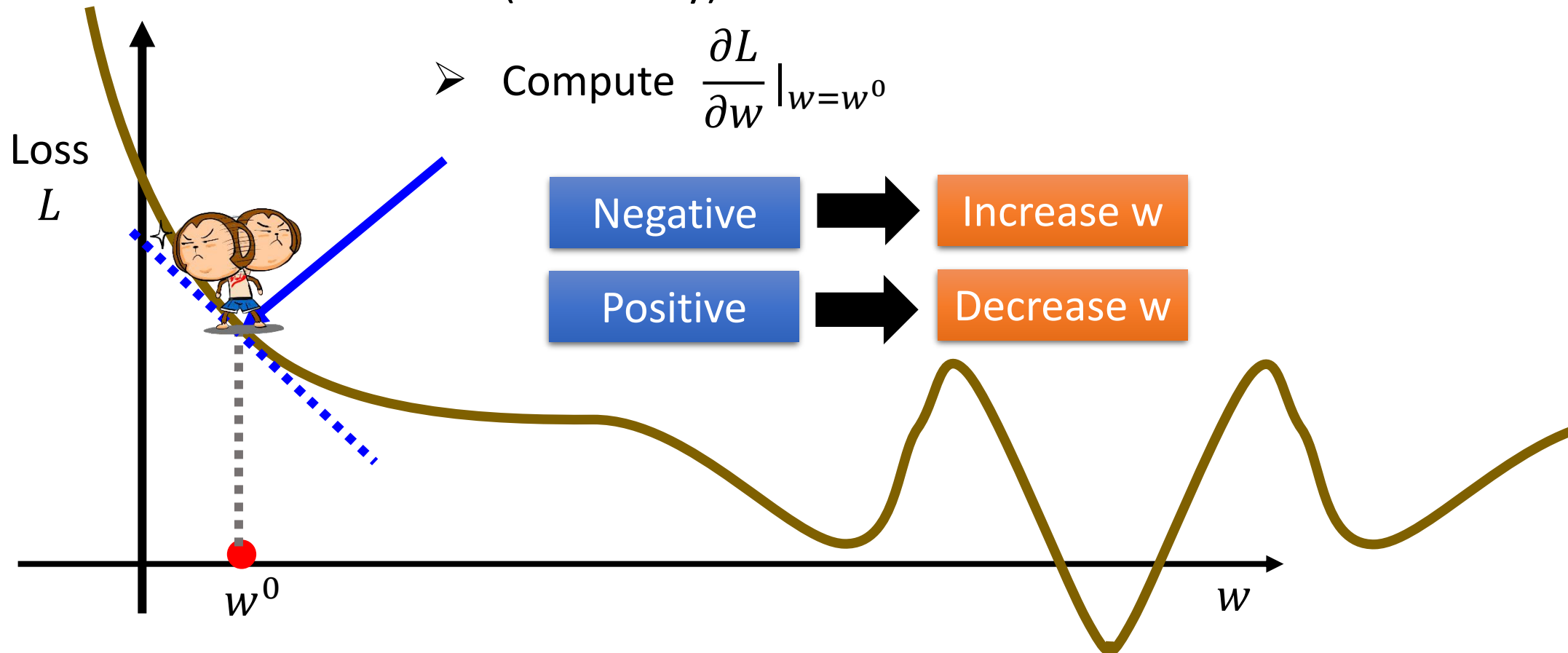


Optimization

$$w^* = \arg \min_w L$$

Gradient Descent

- (Randomly) Pick an initial value w^0
- Compute $\frac{\partial L}{\partial w} \big|_{w=w^0}$



Optimization

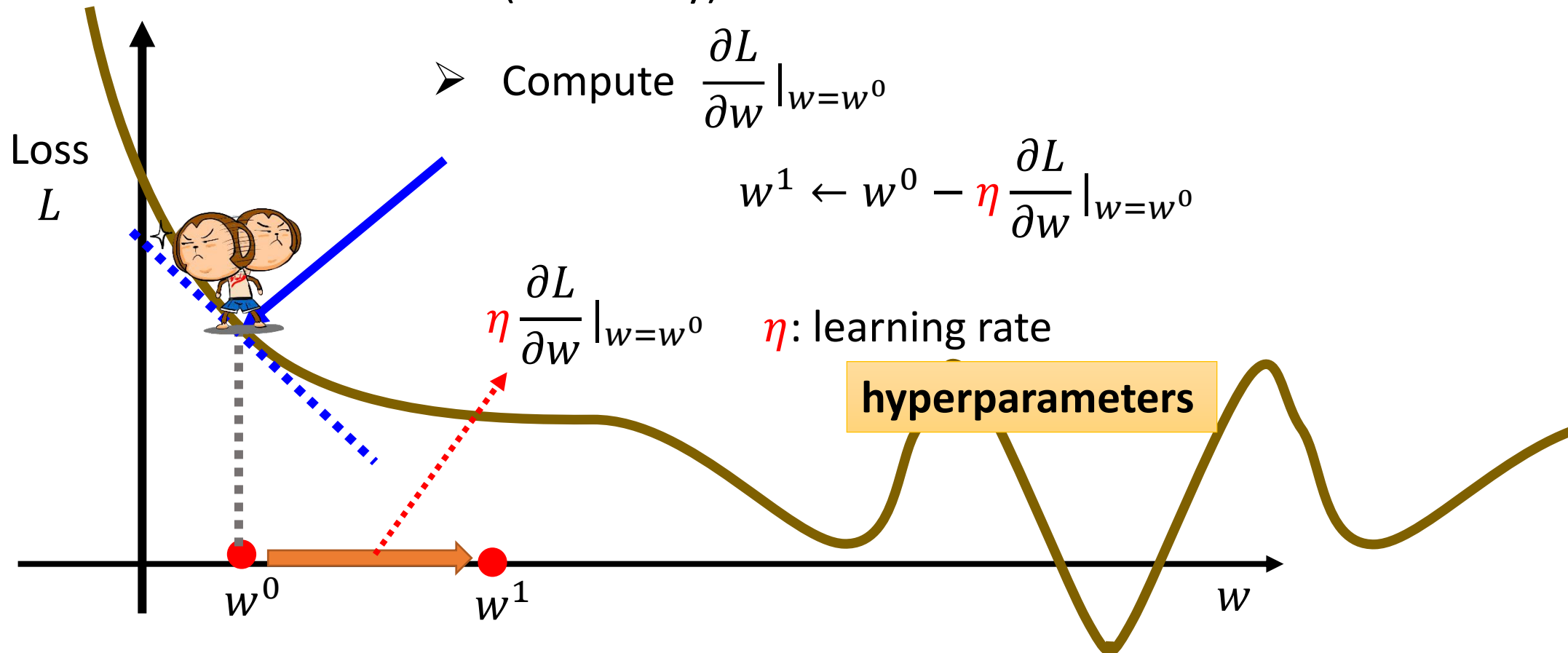
$$w^* = \arg \min_w L$$

Gradient Descent

➤ (Randomly) Pick an initial value w^0

➤ Compute $\frac{\partial L}{\partial w} \big|_{w=w^0}$

$$w^1 \leftarrow w^0 - \eta \frac{\partial L}{\partial w} \big|_{w=w^0}$$



Optimization

$$w^* = \arg \min_w L$$

Gradient Descent

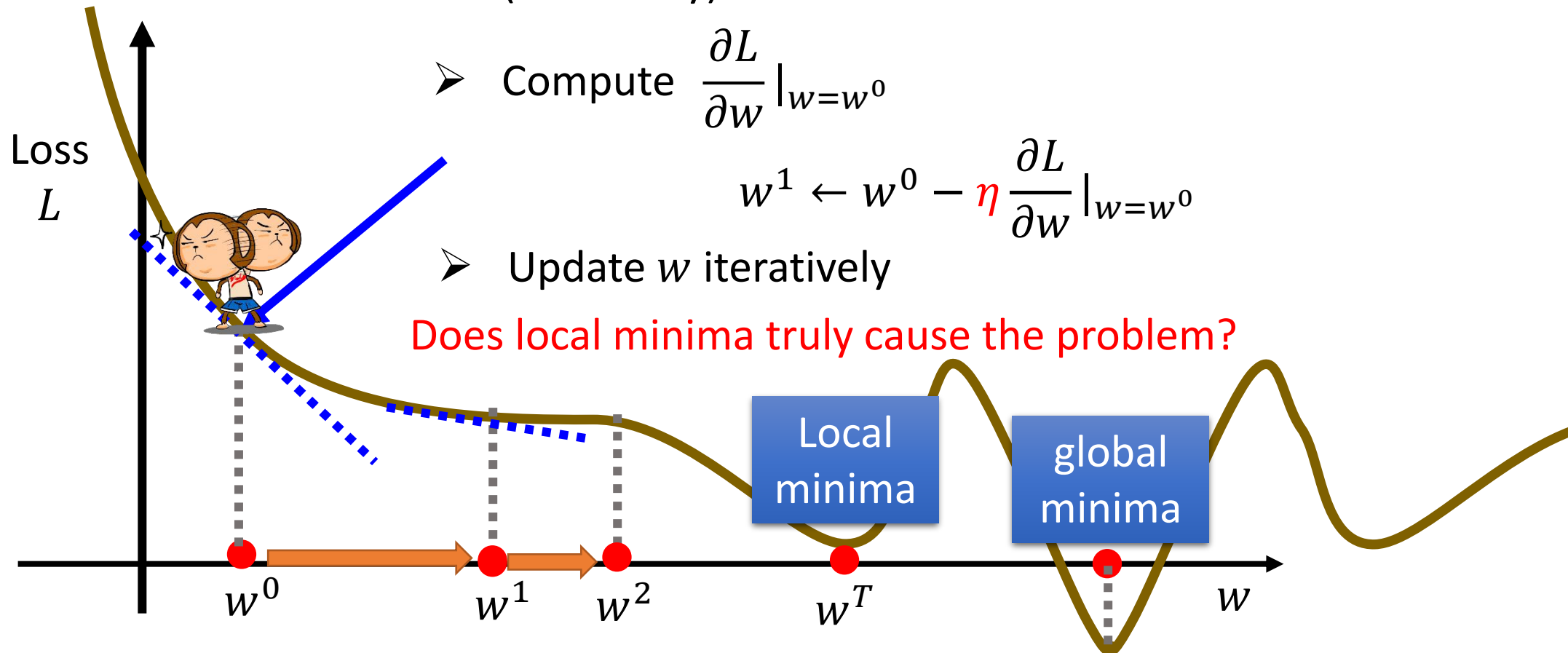
➤ (Randomly) Pick an initial value w^0

➤ Compute $\frac{\partial L}{\partial w} \Big|_{w=w^0}$

$$w^1 \leftarrow w^0 - \eta \frac{\partial L}{\partial w} \Big|_{w=w^0}$$

➤ Update w iteratively

Does local minima truly cause the problem?



Optimization

$$w^*, b^* = \arg \min_{w, b} L$$

- (Randomly) Pick initial values w^0, b^0
- Compute

$$\begin{aligned} \frac{\partial L}{\partial w} \Big|_{w=w^0, b=b^0} \\ \frac{\partial L}{\partial b} \Big|_{w=w^0, b=b^0} \end{aligned}$$


$$w^1 \leftarrow w^0 - \eta \frac{\partial L}{\partial w} \Big|_{w=w^0, b=b^0}$$

$$b^1 \leftarrow b^0 - \eta \frac{\partial L}{\partial b} \Big|_{w=w^0, b=b^0}$$

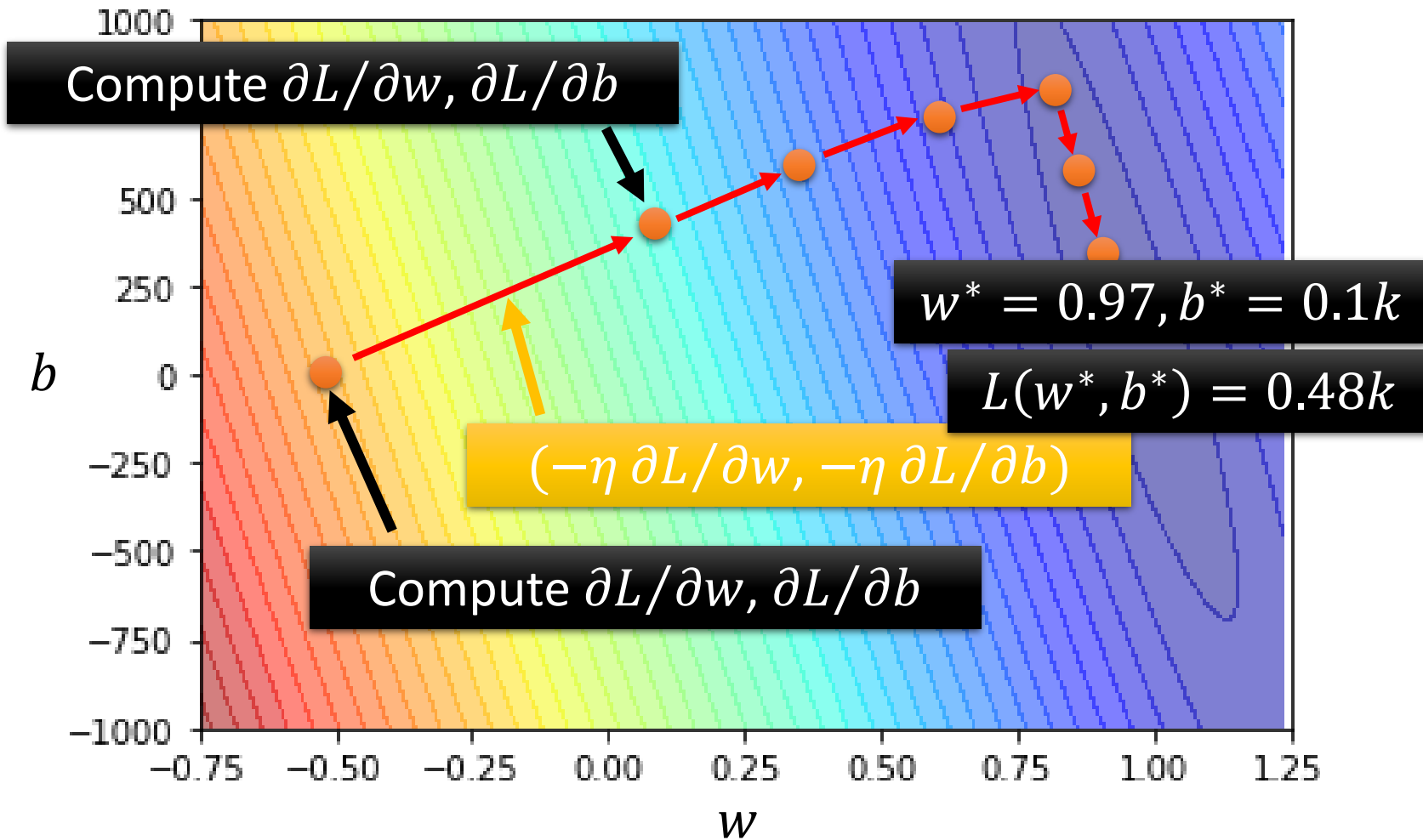
Can be done in one line in most deep learning frameworks

- Update w and b iteratively

Optimization

$$\text{Model } y = b + wx_1$$

$$w^*, b^* = \arg \min_{w, b} L$$



Optimization and Gradient Descent

PYTORCH



Caffe



theano

Summary: Steps in Machine Learning

Step 0: What kind of function do you want to find?

- Know your data: image, text, social networks, reviews
- Know your task: Regression, Classification, Generation

Step 1:
define a set
of function

Deep Learning
SVM
Decision Tree
.....



Step 2:
goodness of
function

Supervised
Reinforcement
.....



Step 3: find
the best
function

Optimization
Gradient Descent
.....